

SA1 Hint Exam — Prof. Kanokporn's Exam Hints

MDS211 Nervous System — 102 Questions · Built From Your Professor's Exact Hint List

Every question below maps directly to one of the topics your professor named. Nothing here is generic filler — it's grounded in her actual slide diagrams (auditory pathway order, VOR wiring, tympanic membrane germ-layer origin, the exact pain classification she teaches) plus a prior cohort's verified exam recall for this same course. Each answer explains why the correct choice is right and why the tempting wrong answers are traps, and "PROF'S EMPHASIS" boxes flag exactly where she's likely to test a specific distinction (like macula sparing) rather than just the topic in general.

Original hint list: Ear — inner ear pathway & Nystagmus, conjugate eye movement, VOR, sound-dampening mechanism, ear pressure adjustment, cochlear base/apex frequency, head rotation, elevator/driving. Eye — eye tissue layers & light absorption/phototransduction, visual pathway lesion terminology, Meyer's loop (emphasis: macula sparing). Ear Dev — tri-germ-layer structures (tympanic membrane), ectoderm/meso/endoderm derivatives. Somatosense — protopathic/epicritic, 4 types of pain, muscle spindle/GTO fiber types, free nerve ending + A/C fibers, tracts, taste pathway.

PART 1 — Ear

Inner Ear Stimulation Pathway & Nystagmus (6 questions)

1. What is the correct order of the auditory pathway, from first-order neuron onward?
 - A. Spiral ganglion (1st) → cochlear nucleus (2nd) → trapezoid body & superior olivary nucleus → lateral lemniscus → inferior colliculus (3rd) → medial geniculate body (4th) → primary auditory cortex
 - B. Cochlear nucleus (1st) → spiral ganglion (2nd) → superior olivary nucleus → inferior colliculus → lateral geniculate body
 - C. Spiral ganglion (1st) → vestibular nucleus (2nd) → medial geniculate body → primary auditory cortex
 - D. Hair cell (1st) → spiral ganglion (2nd) → inferior colliculus (3rd) → primary auditory cortex directly
 - E. Spiral ganglion (1st) → cochlear nucleus (2nd) → lateral geniculate body (3rd) → primary auditory cortex
2. Sound wave transduction: put these structures in the correct order the sound signal actually passes through.
 - A. Ear drum → middle ear ossicles → footplate of stapes (~20x force amplification) → hair cells → spiral ganglia → CN VIII → CNS
 - B. Ear drum → spiral ganglia → ossicles → hair cells → CN VIII
 - C. Oval window → ear drum → ossicles → hair cells → CN VIII
 - D. Ear drum → round window → ossicles → hair cells → CN VIII
 - E. Ossicles → ear drum → hair cells → spiral ganglia → CN VIII
3. Physiologic nystagmus, as part of the vestibulo-ocular reflex, is characterized by which two-phase pattern?
 - A. A slow phase of alternating smooth pursuit in one direction, and a fast (saccadic) phase resetting in the other direction
 - B. Two fast saccadic phases in the same direction
 - C. A single continuous smooth pursuit with no fast phase
 - D. Random, non-patterned eye movement
 - E. A slow phase only, with no reset movement

4. What differentiates pathologic nystagmus from physiologic nystagmus?
- A. Pathologic nystagmus results from damage to one or more vestibular system components — semicircular canals, otolith organs, or the vestibulocerebellum — rather than being a normal VOR response to head movement
 - B. Pathologic nystagmus only occurs during sleep
 - C. Pathologic nystagmus has no fast phase at all
 - D. Pathologic nystagmus is always bilateral and symmetric with no directionality
 - E. There is no meaningful difference between the two
5. A patient's left vestibular nerve is completely severed. Which of the following is TRUE?
- A. This does not produce contralateral (right-sided) nystagmus — the resulting nystagmus/imbalance pattern reflects the LOSS of tonic input from the damaged (left) side, not a mirrored effect on the opposite side
 - B. It produces classic nystagmus with the fast phase toward the intact right side with no other explanation needed
 - C. It has no effect on eye movements at all
 - D. It only affects hearing, never balance
 - E. It causes bilateral, perfectly symmetric vestibular function
6. Damage to the vestibular system (semicircular canals, otolith organs, or vestibulocerebellum) can produce pathologic nystagmus. Which structure is NOT one of these three sources?
- A. Semicircular canals
 - B. Otolith organs (utricle/sacculle)
 - C. Vestibulocerebellum
 - D. Primary auditory cortex
 - E. None — all three listed sources are correct

Conjugate Eye Movement (5 questions)

7. Which cranial nerves are involved in HORIZONTAL conjugate eye movement?
- A. CN III and CN VI
 - B. CN III and CN IV
 - C. CN IV and CN VI
 - D. CN II and CN III
 - E. CN III, IV, and VI all equally for horizontal movement
8. Which cranial nerves are involved in VERTICAL conjugate eye movement?
- A. CN III and CN IV
 - B. CN III and CN VI
 - C. CN IV and CN VI
 - D. CN II and CN VI
 - E. Only CN III
9. What structure carries the signal connecting the cranial nerve nuclei (III, IV, VI) that move the eyeball, coordinating conjugate gaze?
- A. Medial longitudinal fasciculus (MLF)
 - B. Lateral lemniscus
 - C. Medial lemniscus
 - D. Corticospinal tract
 - E. Spinothalamic tract

10. Where is the horizontal gaze center (saccadic center) that generates the fast, voluntary horizontal eye movements (saccades)?

- A. PPRF (Paramedian pontine reticular formation)
- B. Periaqueductal gray
- C. Superior colliculus only
- D. Primary visual cortex
- E. Cerebellar vermis

11. A lesion of the right MLF causes failure of the right eye to adduct on attempted LEFTward conjugate gaze, with nystagmus in the abducting (left) eye. What is this classically called?

- A. Internuclear ophthalmoplegia (INO)
- B. Horizontal gaze palsy
- C. One-and-a-half syndrome
- D. Vertical gaze palsy
- E. Complete CN III palsy

Vestibulo-Ocular Reflex (VOR) (5 questions)

12. During the horizontal vestibulo-ocular reflex, if the head turns LEFT, what happens to the eyes, and what is the nystagmus called?

- A. Both eyes turn RIGHT (slow phase, compensatory), and the resulting nystagmus is named 'to the left' (by its fast/reset phase)
- B. Both eyes turn LEFT with nystagmus to the right
- C. The eyes move in opposite directions from each other (disconjugate)
- D. The eyes do not move at all
- E. Both eyes turn right, and nystagmus is named 'to the right'

13. Which pathway mediates the horizontal VOR? Put it in the correct order.

- A. Lateral crista ampullares (of the horizontal semicircular canal) → vestibular nuclei in the medulla → oculomotor nucleus & abducens nucleus
- B. Utricle → vestibular nuclei in the pons → oculomotor & trochlear nuclei
- C. Saccule → vestibular nuclei in the medulla → abducens nucleus only
- D. Superior crista ampullaris → vestibular nuclei in the pons → oculomotor & trochlear nuclei
- E. Cochlea → superior olivary nucleus → oculomotor nucleus

14. Which pathway mediates the vertical VOR (head moves up → eyes move down, and vice versa)? Put it in the correct order.

- A. Superior and posterior crista ampullares → vestibular nuclei in the pons (superior) → oculomotor nucleus & trochlear nucleus
- B. Lateral crista ampullaris → vestibular nuclei in the medulla → oculomotor & abducens nuclei
- C. Utricle macula → vestibular nuclei in the medulla → trochlear nucleus only
- D. Saccule macula → vestibular nuclei in the pons → abducens nucleus
- E. Cochlea → inferior colliculus → oculomotor nucleus

15. A patient's head moves UP. What happens to the eyes via the vertical VOR?

- A. Eyes move DOWN (compensatory, to keep gaze fixed on the target)
- B. Eyes move UP with the head
- C. Eyes close completely
- D. Eyes move horizontally instead
- E. No reflex occurs for vertical head movement

16. Which nuclei stimulate the medial and lateral rectus muscles for horizontal VOR, based on the vestibular-nuclei-to-extraocular-muscle wiring?

- A. Oculomotor nucleus (medial rectus) and abducens nucleus (lateral rectus)
- B. Trochlear nucleus and abducens nucleus
- C. Oculomotor nucleus and trochlear nucleus
- D. Facial nucleus and abducens nucleus
- E. Vestibular nucleus directly innervates the muscles with no relay

Sound-Dampening Mechanism (5 questions)

17. What is the physiological purpose of the sound-reduction (“glai lod siang”) mechanism involving the stapedius and tensor tympani reflexes?

- A. To inhibit hair cell response to certain frequencies, sharpening perception of other, wanted frequencies — e.g. hearing a friend talk in a noisy room
- B. To permanently destroy hair cells responding to loud sound
- C. To increase the amplitude of ALL sound frequencies equally
- D. To close the Eustachian tube
- E. To generate new hair cells

18. What does contraction of the stapedius muscle (stapedius reflex) mechanically accomplish?

- A. It holds the stapes firm, preventing the oval window from vibrating too vigorously
- B. It holds the malleus firm, adjusting tympanic membrane tension
- C. It opens the Eustachian tube
- D. It contracts the tensor tympani directly
- E. It increases endolymph production

19. What does contraction of the tensor tympani muscle accomplish?

- A. It holds the malleus firm, adjusting the tension of the tympanic membrane so it doesn't vibrate too easily, tuning which frequency range is received well
- B. It holds the stapes firm at the oval window
- C. It has no mechanical effect on the middle ear
- D. It vibrates the round window
- E. It closes the external auditory meatus

20. What is the pathway/circuit driving the sound-reduction (stapedius + tensor tympani) reflex, per your professor's diagram?

- A. Cochlear nucleus → superior olivary nucleus → reticular formation → CN V (innervating tensor tympani) and CN VII (innervating stapedius)
- B. Cochlear nucleus → inferior colliculus → medial geniculate body → CN VIII directly to both muscles
- C. Spiral ganglion → vestibular nuclei → CN III and CN IV
- D. Superior olivary nucleus → cochlear nucleus → CN VII (tensor tympani) and CN V (stapedius)
- E. Reticular formation → cochlear nucleus → CN VIII to both muscles directly

21. A patient has facial nerve (CN VII) damage and develops hyperacusis (sounds seem abnormally loud). Which specific reflex is lost, and why?

- A. The stapedius reflex, because CN VII innervates the stapedius muscle
- B. The tensor tympani reflex, because CN VII innervates the tensor tympani
- C. Both reflexes equally, since CN VII innervates both muscles
- D. Neither reflex is affected by facial nerve damage
- E. The gag reflex

Ear Pressure Equalization (3 questions)

- 22.** What structure equalizes air pressure between the middle ear cavity and the atmosphere (e.g., the 'ear popping' sensation during altitude changes)?
- A. Eustachian (auditory/pharyngotympanic) tube, connecting the middle ear to the nasopharynx
 - B. Round window
 - C. Oval window
 - D. Semicircular canals
 - E. Cochlear duct
- 23.** Blockage of the Eustachian tube (e.g., from an upper respiratory infection) would be expected to cause which type of hearing change?
- A. Conductive hearing changes, from altered middle-ear pressure and reduced tympanic membrane mobility
 - B. Sensorineural hearing loss, from cochlear hair cell death
 - C. No effect on hearing whatsoever
 - D. Immediate permanent deafness
 - E. Loss of vestibular function only
- 24.** Which best distinguishes the round window's function from the oval window's, in the context of pressure within the cochlea?
- A. The oval window is where the stapes pushes fluid IN; the round window bulges OUT to release that pressure wave, allowing continued fluid movement
 - B. Both windows do exactly the same job
 - C. The round window is where sound enters; the oval window releases pressure
 - D. Neither window has anything to do with pressure
 - E. The oval window releases CSF pressure, unrelated to sound

Cochlear Base/Apex Frequency Tuning (4 questions)

- 25.** Which part of the cochlea responds best to HIGH-frequency sound, and why?
- A. The base — the basilar membrane there is short and taut (stiff)
 - B. The apex — the basilar membrane there is short and taut
 - C. The base — the basilar membrane there is long and floppy
 - D. Both base and apex equally
 - E. Neither; frequency is not encoded by cochlear location
- 26.** Which part of the cochlea responds best to LOW-frequency sound, and why?
- A. The apex — the basilar membrane fibers there are long and loose (floppy)
 - B. The base — short and taut fibers
 - C. The apex — short and taut fibers
 - D. The round window region specifically
 - E. The oval window region specifically

27. Why is knowing the cochlea's tonotopic (base=high, apex=low) organization clinically useful when a patient has a middle-ear injury affecting only part of the cochlea?
- A. The pattern of which frequencies are lost on a hearing test can localize WHERE along the cochlea the damage occurred
 - B. It has no clinical usefulness at all
 - C. It only matters for vestibular, not auditory, diagnosis
 - D. All cochlear damage produces identical hearing loss regardless of location
 - E. It only applies to congenital deafness
28. A patient develops high-frequency hearing loss first (as in presbycusis). Based on cochlear tonotopy, which region is most affected?
- A. The base of the cochlea
 - B. The apex of the cochlea
 - C. The vestibule, not the cochlea
 - D. The semicircular canals
 - E. The round window membrane exclusively

Head Rotation — Which Structure Responds (5 questions)

29. Rotating the head horizontally (e.g., shaking the head 'no') primarily stimulates which structure?
- A. The lateral (horizontal) semicircular canal
 - B. The saccule
 - C. The utricle
 - D. The cochlea
 - E. The posterior semicircular canal
30. Flexing the head to touch the chin toward the shoulder (a diagonal/vertical-plane rotational movement) stimulates which structure, per your professor's example?
- A. Posterior semicircular canal
 - B. Lateral semicircular canal
 - C. Utricle
 - D. Saccule
 - E. Cochlea
31. During which condition do the semicircular canals NOT generate extra afferent nerve impulses above baseline?
- A. Constant-velocity rotation (steady-state spinning), because the canals only detect DYNAMIC changes in velocity — acceleration and deceleration — not constant velocity itself
 - B. The onset of head rotation (angular acceleration)
 - C. The moment rotation stops (angular deceleration)
 - D. Any rotational movement whatsoever
 - E. Only during complete stillness
32. Which vestibular nucleus receives input from the posterior semicircular canal AND the saccule?
- A. Inferior vestibular nucleus
 - B. Superior vestibular nucleus
 - C. Lateral vestibular nucleus exclusively
 - D. Medial vestibular nucleus exclusively
 - E. None; these two structures project to different, unrelated nuclei

33. Which vestibular nucleus receives input from the superior/lateral semicircular canals AND the utricle?
- A. Superior vestibular nucleus
 - B. Inferior vestibular nucleus
 - C. Dorsal vestibular nucleus (not a standard nucleus)
 - D. Only the lateral vestibular nucleus
 - E. None of these structures synapse in the brainstem at all

Elevator / Driving — Linear Acceleration (4 questions)

34. Riding in an elevator that suddenly accelerates upward primarily stimulates which structure?
- A. The saccule (vertical linear acceleration)
 - B. The utricle (horizontal linear acceleration)
 - C. The lateral semicircular canal
 - D. The superior semicircular canal
 - E. The cochlea
35. A car accelerating forward from a stop primarily stimulates which structure?
- A. The utricle (horizontal linear acceleration)
 - B. The saccule (vertical linear acceleration)
 - C. The posterior semicircular canal
 - D. The lateral semicircular canal
 - E. The cochlea
36. What are the otolith organs (utricle and saccule) structurally specialized with, that gives them the inertial mass needed to detect linear acceleration?
- A. Otoconia (calcium carbonate crystals) embedded in an otolithic membrane overlying the hair cell stereocilia
 - B. A cupula with no crystals, like the semicircular canals
 - C. A basilar membrane
 - D. A tectorial membrane only, with no crystals
 - E. No specialized structure; they work identically to the semicircular canals
37. Walking forward at a steady pace (repeated small forward accelerations/decelerations with each step) is most associated with stimulation of which structure(s), based on horizontal linear movement?
- A. Utricle (macula)
 - B. Saccule (macula)
 - C. Posterior semicircular canal
 - D. Superior semicircular canal
 - E. Cochlea

PART 2 — Eye

Eye Tissue Layers: Light Absorption & Phototransduction (7 questions)

- 38.** The eye is composed of three tissue layers (tunics). Which layer is the middle layer, and what does it consist of?
- A. Tunica vasculosa (uvea) — choroid, ciliary body, and iris
 - B. Tunica fibrosa — sclera and cornea
 - C. Retina — pigmented and neural layers
 - D. Tunica vasculosa — sclera and cornea only
 - E. There are only two tissue layers in the eye
- 39.** Which layer of the eye is primarily responsible for ABSORBING stray light, preventing internal scatter/reflection that would blur the image?
- A. Choroid (heavily pigmented, part of the tunica vasculosa)
 - B. Cornea
 - C. Sclera
 - D. Retina (specifically the neural layer)
 - E. Lens
- 40.** Which structure actually converts light energy into a nerve impulse (phototransduction)?
- A. Photoreceptor cells (rods and cones) in the retina, via their photopigments
 - B. The choroid
 - C. The cornea
 - D. The sclera
 - E. The ciliary body
- 41.** Put the phototransduction cascade events in the CORRECT order, from light hitting the photopigment to the final electrical change.
- A. Structural change of the photopigment → activation of transducin → activation of phosphodiesterase → decreased intracellular cGMP → closure of Na⁺ channel
 - B. Closure of Na⁺ channel → activation of transducin → structural change of photopigment → decreased cGMP → activation of phosphodiesterase
 - C. Activation of phosphodiesterase → structural change of photopigment → activation of transducin → closure of Na⁺ channel → decreased cGMP
 - D. Decreased cGMP → structural change of photopigment → closure of Na⁺ channel → activation of transducin → activation of phosphodiesterase
 - E. Structural change of photopigment → closure of Na⁺ channel → activation of transducin → activation of phosphodiesterase → decreased cGMP
- 42.** Unlike most sensory receptors, photoreceptors respond to their stimulus (light) with which membrane change?
- A. Hyperpolarization (light closes Na⁺ channels, reducing the dark current)
 - B. Depolarization, exactly like other sensory receptors
 - C. No membrane change occurs at all
 - D. Complete depolarization block with no graded response
 - E. An all-or-none action potential directly at the photoreceptor
- 43.** Which retinal cell layer contains the cell bodies of the photoreceptors themselves?
- A. Outer nuclear layer
 - B. Inner nuclear layer
 - C. Ganglion cell layer
 - D. Inner plexiform layer
 - E. Nerve fiber layer

44. Correctly order the retinal layers from the rod/cone layer toward the optic nerve fiber layer.
- A. Rods/cones → outer nuclear layer → inner nuclear layer → ganglion cell layer → optic nerve fiber layer
 - B. Rods/cones → inner nuclear layer → outer nuclear layer → ganglion cell layer → optic nerve fiber layer
 - C. Rods/cones → ganglion cell layer → inner nuclear layer → optic nerve fiber layer
 - D. Rods/cones → outer nuclear layer → ganglion cell layer → inner nuclear layer → optic nerve fiber layer
 - E. Optic nerve fiber layer → ganglion cell layer → inner nuclear layer → outer nuclear layer → rods/cones (reverse order)

Visual Pathway Lesions — Specific Terminology (10 questions)

45. A lesion of the OPTIC NERVE (before the chiasm) causes which type of visual field defect?
- A. Monocular blindness (or a monocular field defect) in the eye on the same side as the lesion
 - B. Bitemporal hemianopia
 - C. Binasal hemianopia
 - D. A homonymous field defect affecting both eyes equally
 - E. No visual deficit at all
46. A lesion at the MIDLINE of the optic chiasm (e.g., from a pituitary tumor pressing upward) classically causes which visual field defect?
- A. Bitemporal heteronymous hemianopia
 - B. Binasal heteronymous hemianopia
 - C. Right homonymous hemianopia
 - D. Left homonymous hemianopia
 - E. A quadrantanopia
47. A lesion at the LATERAL part of the optic chiasm (e.g., bilateral compressive lesions from the sides) classically causes which visual field defect?
- A. Binasal heteronymous hemianopia
 - B. Bitemporal heteronymous hemianopia
 - C. A homonymous hemianopia
 - D. Complete bilateral blindness
 - E. No field defect, since lateral fibers don't carry visual information
48. A lesion of the OPTIC TRACT (just after the chiasm) causes which visual field defect?
- A. Homonymous hemianopia (complete half-field loss in both eyes, on the side opposite the lesion), WITHOUT macular sparing
 - B. Monocular blindness
 - C. Bitemporal hemianopia
 - D. A quadrantanopia only
 - E. Homonymous hemianopia WITH macular sparing
49. A lesion of the OPTIC RADIATION (complete, affecting both the parietal and temporal fibers) causes which visual field defect?
- A. Homonymous hemianopia (on the side opposite the lesion)
 - B. A quadrantanopia only, never a full hemianopia
 - C. Bitemporal hemianopia
 - D. Monocular blindness
 - E. Binasal hemianopia

- 50.** A SELECTIVE lesion of the optic radiation fibers (i.e., only part of it, such as Meyer's loop alone) causes which visual field defect?
- A. A quadrantanopia (loss of one specific quadrant of the visual field)
 - B. A complete homonymous hemianopia
 - C. Bitemporal hemianopia
 - D. Monocular blindness
 - E. No visual deficit
- 51.** Which structure, if damaged, produces a homonymous hemianopia WITH macular sparing — the pattern your professor specifically emphasizes?
- A. The occipital (visual) cortex — because the macular region has a dual blood supply (both MCA and PCA collateral flow), it's often spared even when the rest of the visual cortex is infarcted
 - B. The optic nerve
 - C. The optic chiasm (midline)
 - D. The optic tract (this classically does NOT spare the macula)
 - E. The lens
- 52.** A patient has a RIGHT occipital lobe infarct (PCA territory). What visual field defect is expected?
- A. Left homonymous hemianopia WITH macular sparing
 - B. Right homonymous hemianopia with macular sparing
 - C. Left homonymous hemianopia WITHOUT macular sparing
 - D. Bitemporal hemianopia
 - E. Right monocular blindness
- 53.** A lesion of the RIGHT Meyer's loop (temporal lobe portion of the optic radiation) causes which specific visual field defect?
- A. Left homonymous SUPERIOR quadrantanopia (pie in the sky)
 - B. Right homonymous inferior quadrantanopia
 - C. Right homonymous superior quadrantanopia
 - D. Left homonymous inferior quadrantanopia
 - E. Left homonymous hemianopia with macular sparing
- 54.** A lesion of the PARIETAL lobe fibers of the optic radiation (superior fibers, NOT Meyer's loop) causes which visual field defect?
- A. A contralateral INFERIOR quadrantanopia (pie on the floor)
 - B. A contralateral SUPERIOR quadrantanopia
 - C. A complete homonymous hemianopia always
 - D. Bitemporal hemianopia
 - E. No visual field defect

Meyer's Loop & Macula Sparing — Prof's Emphasis (5 questions)

55. Why does the OPTIC TRACT lesion NOT spare the macula, while an OCCIPITAL CORTEX lesion classically DOES?

- A. The optic tract is a single, compact bundle with a single blood supply and no built-in redundancy for any part of the visual field, including the macula; the occipital cortex's macular representation specifically benefits from dual (MCA + PCA) blood supply not available further upstream
- B. There is no real difference; both always spare the macula equally
- C. The optic tract has a larger macular representation than the cortex
- D. Macular fibers physically bypass the optic tract entirely
- E. The occipital cortex has no blood supply to the macula at all

56. A 65-year-old patient has a stroke and is found to have a complete left homonymous hemianopia, but central (macular) vision is preserved in both eyes. Where does this localize the lesion, and to which side?

- A. Right occipital cortex (likely right PCA territory)
- B. Right optic tract
- C. Left occipital cortex
- D. Optic chiasm
- E. Left optic nerve

57. A lesion of the right optic tract (as opposed to the right occipital cortex) causes left homonymous hemianopia. What KEY difference distinguishes this from a right occipital cortex lesion on exam?

- A. The optic tract lesion does NOT spare the macula, while the occipital cortex lesion classically DOES
- B. The optic tract lesion always spares the macula, while the cortex lesion never does
- C. There is no distinguishing feature; the exam finding is identical
- D. The optic tract lesion causes bitemporal, not homonymous, loss
- E. Only the cortex lesion causes any visual field defect at all

58. Which best describes why 'macula sparing' is clinically valuable, beyond just being a testable fact?

- A. It helps localize a homonymous hemianopia to the cortex (implying a vascular/PCA stroke mechanism) rather than to the optic tract (implying a different, often compressive or different-vascular-territory process), guiding further workup
- B. It has no real clinical value beyond trivia
- C. It only matters for choosing glasses prescriptions
- D. It determines whether the patient needs cataract surgery
- E. It only applies to congenital, not acquired, visual field defects

59. Combining what you know: a patient has a LEFT homonymous INFERIOR quadrantanopia. Which structure is most likely damaged, and on which side?

- A. Right parietal lobe optic radiation fibers (carrying the contralateral inferior field)
- B. Right Meyer's loop (temporal lobe)
- C. Left parietal lobe optic radiation fibers
- D. Left Meyer's loop
- E. Right occipital cortex, complete

PART 3 — Ear Development

Tri-Germ-Layer Structures (Tympanic Membrane) (4 questions)

- 60.** The tympanic membrane (eardrum) is a rare structure that develops from all THREE germ layers. Match each layer to the tympanic membrane component it forms.
- A. Outer layer = ectoderm (from the 1st pharyngeal cleft); middle layer = mesoderm (fibrous connective tissue); inner layer = endoderm (from the 1st pharyngeal pouch)
 - B. Outer layer = endoderm; middle layer = ectoderm; inner layer = mesoderm
 - C. All three layers of the tympanic membrane are ectoderm only
 - D. Outer layer = mesoderm; middle layer = endoderm; inner layer = ectoderm
 - E. The tympanic membrane has no mesodermal contribution at all
- 61.** Which structure marks the boundary between the ectoderm-derived external ear and the endoderm-derived middle ear during development?
- A. The tympanic membrane
 - B. The pinna (auricle)
 - C. The cochlea
 - D. The vestibule
 - E. The semicircular canals
- 62.** Which ear structures develop from the first pharyngeal CLEFT (ectoderm)?
- A. The external auditory meatus (and contributes to the outer layer of the tympanic membrane)
 - B. The ossicles (malleus, incus, stapes)
 - C. The tympanic cavity and Eustachian tube
 - D. The membranous labyrinth (cochlea, semicircular canals)
 - E. The facial nerve
- 63.** Which ear structures develop from the first pharyngeal POUCH (endoderm)?
- A. The tympanic cavity (middle ear space) and auditory (Eustachian) tube, plus the inner epithelial layer of the tympanic membrane
 - B. The external auditory meatus
 - C. The pinna cartilage
 - D. The cochlear duct
 - E. The vestibular nerve

Ectoderm / Mesoderm / Endoderm — General Derivatives (6 questions)

- 64.** In general terms, what does surface ECTODERM give rise to, among structures relevant to the special senses?
- A. Lens of the eye, corneal epithelium, external ear/auditory meatus, olfactory epithelium, and (with endoderm/mesoderm) the outer tympanic membrane layer
 - B. The choroid, sclera, and corneal endothelium
 - C. The middle ear ossicles
 - D. The retina and optic nerve
 - E. The inner ear membranous labyrinth
- 65.** What does NEUROectoderm give rise to, among special-sense structures?
- A. The retina, optic nerve, and the otic vesicle/membranous labyrinth (inner ear) — both being direct outgrowths of neural tissue
 - B. The lens and corneal epithelium
 - C. The middle ear ossicles
 - D. The pinna cartilage
 - E. The external auditory meatus

66. What does MESODERM contribute to the special sense organs?
- A. Connective tissue and vascular contributions — e.g., the fibrous/vascular coats of the eye, the middle layer of the tympanic membrane, and (with neural crest) much of the periocular/perilabyrinthine mesenchyme
 - B. The retina exclusively
 - C. The lens exclusively
 - D. The membranous labyrinth exclusively
 - E. The olfactory epithelium
67. What does ENDODERM contribute to the ear, specifically?
- A. The middle ear cavity (tympanic cavity) and auditory tube, plus the inner layer of the tympanic membrane, all via the first pharyngeal pouch
 - B. The external auditory meatus
 - C. The cochlear hair cells
 - D. The vestibular ganglion
 - E. The lens
68. Which eye structures are neural CREST derivatives (as opposed to surface ectoderm, neuroectoderm, or mesoderm proper)?
- A. Corneal endothelium, choroid, sclera, and much of the corneal stroma
 - B. The lens
 - C. The retina
 - D. The optic nerve
 - E. The external auditory meatus
69. True or false framing: since the tympanic membrane draws on all three germ layers, which single statement correctly captures why this is considered a notable/unusual embryological fact worth testing?
- A. Most individual adult structures are derived from predominantly ONE germ layer; the tympanic membrane is a rare example of a single structure requiring coordinated contribution from all three (ectoderm, mesoderm, endoderm), reflecting its position exactly at the ectoderm/endoderm boundary between external and middle ear
 - B. All structures in the body draw equally from all three germ layers, so this is not actually unusual
 - C. The tympanic membrane is entirely mesodermal and this is a trick question
 - D. Germ layer origin has no bearing on adult structure at all
 - E. This fact applies to every membrane in the body equally

PART 4 — Somatosense

Protopathic vs. Epicritic Sensation (5 questions)

70. Which sensations are classified as PROTOPATHIC (affective) sensation, per your professor's table?
- A. Pain, hot & cold (temperature extremes), crude/gross touch
 - B. Fine/light touch, vibration, two-point discrimination, proprioception
 - C. Only pain
 - D. Only proprioception
 - E. Vibration and two-point discrimination

- 71.** Which sensations are classified as DISCRIMINATIVE (epicritic) sensation?
- A. Fine/light touch, pressure, vibration, two-point discrimination, and proprioceptive sense
 - B. Pain and hot/cold only
 - C. Only crude touch
 - D. Only pain
 - E. Only pressure
- 72.** Structurally, how do protopathic and epicritic receptors/fibers differ?
- A. Protopathic: no receptor covering, nonmyelinated axon, branching nerve endings. Epicritic: receptor has a covering/capsule, myelinated axon, NO branching of the nerve ending
 - B. Protopathic: myelinated, capsulated, no branching. Epicritic: nonmyelinated, no capsule, branching
 - C. Both have identical structure with no meaningful difference
 - D. Protopathic fibers are always thicker than epicritic fibers
 - E. Epicritic receptors are always free nerve endings
- 73.** Within the dorsal root of a spinal nerve, epicritic and protopathic fibers enter via different divisions. Which is correct?
- A. Epicritic fibers (large, myelinated) enter the MEDIAL division of the posterior rootlet; protopathic fibers (small, nonmyelinated) enter the LATERAL division
 - B. Epicritic fibers enter laterally; protopathic fibers enter medially
 - C. Both enter via the exact same division with no separation
 - D. Epicritic fibers do not enter the dorsal root at all
 - E. Protopathic fibers only enter via the ventral root
- 74.** Within the dorsal root/spinal cord entry zone, sensory fiber types are also organized medial-to-lateral by MODALITY. Which is the correct medial-to-lateral order?
- A. Unconscious proprioception → conscious proprioception → touch/pressure → pain & temperature
 - B. Pain & temperature → touch/pressure → conscious proprioception → unconscious proprioception
 - C. Touch/pressure → pain & temperature → proprioception (both types)
 - D. All modalities enter at the exact same mediolateral position
 - E. Pain & temperature → unconscious proprioception → conscious proprioception → touch/pressure

The Four Types of Pain (6 questions)

- 75.** Your professor's lecture describes FOUR types of pain. Which of the following correctly lists all four?
- A. Superficial pain (with fast and slow subtypes), Deep pain, Visceral pain, and Referred pain
 - B. Fast pain, Slow pain, Chronic pain, and Acute pain only
 - C. Only Fast and Slow pain exist as categories
 - D. Somatic pain, Neuropathic pain, Psychogenic pain, and Idiopathic pain
 - E. Superficial, Deep, Central, and Peripheral pain
- 76.** SUPERFICIAL pain arises from which germ layer, and what are its two subtypes?
- A. Ectoderm; fast pain (myelinated A-delta fibers, sharp/well-localized) and slow pain (unmyelinated C fibers, dull/poorly localized)
 - B. Mesoderm; fast and slow subtypes
 - C. Endoderm; only one subtype exists
 - D. Neural crest; no subtypes exist
 - E. Mesoderm; visceral and referred subtypes

77. DEEP pain arises from which germ layer and which structures, and what is its characteristic clinical presentation?

- A. Mesoderm; from muscle, tendon, joint, periosteum, and ligament (e.g., ischemia/muscle cramp); poorly localized, deep/dull ache over a wide area, often WITH autonomic symptoms (nausea, sweating, low BP, slow HR)
- B. Ectoderm; from skin only; always sharp and precisely localized
- C. Endoderm; from the gut only
- D. Neural crest; from melanocytes
- E. Mesoderm; always well-localized with no autonomic component

78. VISCERAL pain has which characteristic features, per your lecture?

- A. Triggered by mechanical distension or ischemia; few receptors so poorly localized; conducted via CN VII, IX, X and the autonomic nervous system to the thalamus and then postcentral gyrus (areas 3,1,2)
- B. Always sharp and precisely localized to a single dermatome
- C. Carried exclusively by somatic A-beta fibers
- D. Never involves the autonomic nervous system
- E. Only occurs in skeletal muscle

79. What is REFERRED pain, and what theory explains it?

- A. Somatic pain felt in a body region that shares the same dermatome as an internal organ experiencing visceral or deep pain, explained by convergence theory and facilitation theory
- B. Pain that travels from one limb to the opposite limb via the corpus callosum
- C. Pain that has no identifiable cause or mechanism
- D. A type of pain exclusive to the face
- E. Pain caused directly by nerve compression at the site felt

80. Which theory of referred pain proposes that visceral and somatic sensory afferents converge onto the SAME second-order (or higher) neuron in the CNS, causing the brain to misattribute the pain's true source?

- A. Convergence theory
- B. Facilitation theory (exclusively, with no role for convergence)
- C. Gate-control theory
- D. Specificity theory
- E. Pattern theory

Muscle Spindle & Golgi Tendon Organ — Fiber Types (6 questions)

81. Which afferent fiber type detects stretch of the muscle SPINDLE (intrafusal fibers)?

- A. Ia afferent fiber
- B. Ib afferent fiber
- C. II afferent fiber only, never Ia
- D. A-delta fiber
- E. C fiber

82. Which afferent fiber type detects stretch/tension of the GOLGI TENDON ORGAN?

- A. Ib afferent fiber
- B. Ia afferent fiber
- C. II afferent fiber
- D. A-beta fiber
- E. C fiber

- 83.** Regarding the Achilles tendon reflex, which statement is correct?
- A. The receptor is the muscle spindle (not the Golgi tendon organ, despite the tendon in the name)
 - B. The afferent fiber is the Golgi tendon organ
 - C. The effector is the hamstring muscle
 - D. The reflex center is at L1-L2
 - E. This reflex uses Ib afferents exclusively
- 84.** Regarding nuclear bag and nuclear chain intrafusal fibers within the muscle spindle, which correctly describes their afferent innervation?
- A. Nuclear bag fibers carry both static and dynamic responses (Ia primary afferent, with dynamic bag fibers giving the dynamic component); nuclear chain fibers carry mainly static response (Ia and II afferents)
 - B. Nuclear bag fibers carry only Ib afferents
 - C. Nuclear chain fibers carry only Ia afferents with no II component
 - D. Neither fiber type is innervated by group Ia afferents
 - E. Both fiber types are purely efferent, carrying no sensory information
- 85.** What is the function of GAMMA motor neurons in relation to the muscle spindle?
- A. They maintain the sensitivity of the muscle spindle (by adjusting intrafusal fiber tension) during muscle shortening, so the spindle keeps detecting stretch even as the whole muscle contracts
 - B. They directly innervate and contract extrafusal (main, force-producing) muscle fibers
 - C. They maintain the sensitivity of the Golgi tendon organ
 - D. They prevent the muscle from producing too much force
 - E. They have no functional role in normal physiology
- 86.** When dynamic gamma motor neurons are co-activated together with alpha motor neurons during a voluntary contraction, what is the effect on Ia afferent firing, compared to alpha activation alone?
- A. MORE Ia impulses are generated (gamma coactivation keeps the spindle taut and responsive, preventing it from falling silent as the muscle shortens)
 - B. FEWER Ia impulses are generated
 - C. Ia inhibition occurs completely, silencing the afferent
 - D. Clonus always results
 - E. The muscle fails to contract at all

Free Nerve Ending + A-delta / C Fibers (4 questions)

- 87.** Free nerve endings are found in the clefts between epidermal cells and convey which type of information?
- A. Nociceptive (pain) and thermal information related to cellular injury
 - B. Only fine, discriminative touch
 - C. Only proprioceptive information
 - D. Only vibration sense
 - E. Only two-point discrimination
- 88.** A pinprick to the skin activates which fiber type FIRST, producing fast/first pain?
- A. A-delta fibers
 - B. C fibers
 - C. A-beta fibers
 - D. A-alpha fibers
 - E. B fibers

89. A dull, persistent ache that follows a muscle pull results from activation of which fiber type?
- A. C fibers
 - B. A-delta fibers
 - C. A-beta fibers
 - D. A-alpha fibers
 - E. B fibers
90. Which chemical mediators are associated with activating/sensitizing free nerve ending nociceptors, per your lecture?
- A. Histamine, substance P, prostaglandin, and K⁺
 - B. Only acetylcholine
 - C. Only dopamine
 - D. Only GABA
 - E. Only serotonin

Somatosensory Tracts (7 questions)

91. Which THREE major ascending sensory pathways carry general somatic afferent (GSA) information (touch, pressure, vibration, pain, temperature, stretch/position)?
- A. (1) Anterolateral system (anterior + lateral spinothalamic tracts), (2) Dorsal column-medial lemniscus system (fasciculus gracilis + cuneatus), (3) Somatosensory-to-cerebellum pathways (ventral/dorsal spinocerebellar, cuneocerebellar)
 - B. Only the corticospinal tract
 - C. Only the rubrospinal and vestibulospinal tracts
 - D. Only the anterolateral system, with no dorsal column contribution
 - E. Only descending motor tracts, since GSA is purely a motor concept
92. Which tract carries CONSCIOUS proprioception, fine/discriminative touch, and vibration sense?
- A. Dorsal column-medial lemniscus pathway (fasciculus gracilis + cuneatus)
 - B. Lateral spinothalamic tract
 - C. Ventral spinocerebellar tract
 - D. Dorsal spinocerebellar tract
 - E. Anterior spinothalamic tract
93. Which tract carries pain and temperature sensation, and where does it cross the midline?
- A. Lateral spinothalamic tract, crossing within 1-2 segments of entering the cord (via the anterior white commissure)
 - B. Fasciculus gracilis, crossing in the medulla
 - C. Dorsal spinocerebellar tract, uncrossed
 - D. Corticospinal tract, crossing at the pyramidal decussation
 - E. Fasciculus cuneatus, crossing in the medulla
94. Which somatosensory tract synapses at Clarke's nucleus, carrying UNCONSCIOUS proprioception from the lower limb to the cerebellum?
- A. Dorsal spinocerebellar tract
 - B. Ventral spinocerebellar tract
 - C. Lateral spinothalamic tract
 - D. Fasciculus gracilis
 - E. Corticospinal tract

95. Regarding touch sensation specifically, which statement correctly identifies the SECOND-order neuron's synapse location?

- A. The second-order neuron for touch (via the dorsal column pathway) synapses in the cuneate (or gracile) nucleus of the brainstem/medulla
- B. The second-order neuron synapses directly in the thalamus
- C. The second-order neuron synapses in the primary somatosensory cortex
- D. There is no second-order neuron for touch
- E. The second-order neuron synapses in the dorsal root ganglion

96. For touch sensation, in which structure does the FOURTH-order neuron reside?

- A. Primary somatosensory (somesthetic) area of the cerebral cortex
- B. The thalamus
- C. The cuneate nucleus
- D. The dorsal root ganglion
- E. There is no fourth-order neuron for touch

97. Which tract carries pain and temperature information from the FACE (via the spinal trigeminal nucleus), functionally analogous to the spinothalamic tract for the body?

- A. Ventral trigeminothalamic tract
- B. Dorsal column-medial lemniscus pathway
- C. Dorsal spinocerebellar tract
- D. Corticospinal tract
- E. Rubrospinal tract

Taste (Gustatory) Pathway (5 questions)

98. What is the correct taste (gustatory) pathway from receptor to cortex?

- A. Taste receptor cells (CN VII, IX, X) → Geniculate/Petrosal/Inferior(nodose) ganglia → Solitary nucleus (rostral/upper part) → Solitary tract → VPM of thalamus → Gustatory area of cerebral cortex (area 43, near the midline)
- B. Taste receptor cells → Spiral ganglion → Cochlear nucleus → Medial geniculate body → Primary auditory cortex
- C. Taste receptor cells → Trigeminal ganglion → Spinal trigeminal nucleus → VPL of thalamus → Primary somatosensory cortex
- D. Taste receptor cells → Vestibular ganglion → Vestibular nuclei → Thalamus → Vestibular cortex
- E. Taste receptor cells → Dorsal root ganglion → Dorsal column → VPL → Postcentral gyrus

99. Which ganglia house the first-order taste neurons for CN VII, IX, and X respectively?

- A. CN VII = Geniculate ganglion; CN IX = Petrosal ganglion; CN X = Inferior (nodose) ganglion
- B. CN VII = Petrosal; CN IX = Geniculate; CN X = Trigeminal
- C. All three use the same single ganglion
- D. CN VII = Spiral ganglion; CN IX = Vestibular ganglion; CN X = Nodose ganglion
- E. CN VII = Nodose; CN IX = Geniculate; CN X = Petrosal

100. Which part of the solitary nucleus specifically handles TASTE information, as opposed to general visceral/cardiopulmonary information?

- A. The rostral (upper) part of the solitary nucleus
- B. The caudal (lower) part of the solitary nucleus
- C. Taste and visceral information are processed in identical locations with no separation
- D. The solitary nucleus has no role in taste at all
- E. The lateral part exclusively, with no rostro-caudal distinction

101. What is the medical term for an unpleasant or distorted taste sensation?

- A. Dysgeusia
- B. Dysphagia
- C. Dysphonia
- D. Dysarthria
- E. Dysmetria

102. Which stem cell type regenerates sensory receptor cells in BOTH the tongue (taste buds) and the olfactory epithelium?

- A. Basal cells
- B. Satellite cells
- C. Schwann cells
- D. Merkel cells
- E. Stem keratinocytes

Answer Key & Full Explanations

102 Questions · Organized by Hint Topic

PART 1 — Ear

Inner Ear Stimulation Pathway & Nystagmus (6 questions)

Q1. What is the correct order of the auditory pathway, from first-order neuron onward?

✓ **Correct: A. Spiral ganglion (1st) → cochlear nucleus (2nd) → trapezoid body & superior olivary nucleus → lateral lemniscus → inferior colliculus (3rd) → medial geniculate body (4th) → primary auditory cortex**

This is the exact auditory pathway sequence from your lecture: hair cells transduce sound → spiral ganglion (1st-order bipolar neurons) → cochlear nucleus (2nd-order) → trapezoid body/superior olivary complex (first major bilateral relay) → lateral lemniscus → inferior colliculus (3rd-order) → medial geniculate body of the thalamus (4th-order) → primary auditory cortex.

✗ *E is wrong: Lateral geniculate body is the THALAMIC RELAY FOR VISION, not hearing — hearing uses the MEDIAL geniculate body. This exact swap is a favorite trap.*

PROF'S EMPHASIS: This exact chain (spiral ganglion → cochlear nucleus → trapezoid body/superior olivary → lateral lemniscus → inferior colliculus → medial geniculate body → cortex) is straight from your professor's slide — memorize it as a single unbroken chain, since she tests both the full order AND which structure is 'Nth order.'

Q2. Sound wave transduction: put these structures in the correct order the sound signal actually passes through.

✓ **Correct: A. Ear drum → middle ear ossicles → footplate of stapes (~20x force amplification) → hair cells → spiral ganglia → CN VIII → CNS**

This is the mechanical sound-transduction chain exactly as taught: sound wave → tympanic membrane → middle ear ossicles → stapes footplate (amplifies force ~20-fold at the oval window) → cochlear hair cells → spiral ganglia (1st-order) → CN VIII → CNS.

PROF'S EMPHASIS: Your professor's slide gives the exact 20x number for stapes footplate amplification — that specific multiplier is a very plausible fill-in-the-blank or numeric-recall target.

Q3. Physiologic nystagmus, as part of the vestibulo-ocular reflex, is characterized by which two-phase pattern?

✓ **Correct: A. A slow phase of alternating smooth pursuit in one direction, and a fast (saccadic) phase resetting in the other direction**

Physiologic (VOR-driven) nystagmus alternates a slow phase (smooth pursuit, driven by the vestibular reflex trying to keep gaze fixed) with a fast saccadic phase that resets the eyes in the opposite direction — nystagmus is conventionally NAMED by the direction of this fast phase.

Q4. What differentiates pathologic nystagmus from physiologic nystagmus?

✓ **Correct: A. Pathologic nystagmus results from damage to one or more vestibular system components — semicircular canals, otolith organs, or the vestibulocerebellum — rather than being a normal VOR response to head movement**

Physiologic nystagmus is a normal, expected VOR response to head movement; pathologic nystagmus instead signals damage somewhere in the vestibular system (canals, otoliths, or vestibulocerebellum) and can occur even without head movement.

Q5. A patient's left vestibular nerve is completely severed. Which of the following is TRUE?

✓ **Correct: A. This does not produce contralateral (right-sided) nystagmus — the resulting nystagmus/imbalance pattern reflects the LOSS of tonic input from the damaged (left) side, not a mirrored effect on the opposite side**

Cutting one vestibular nerve removes tonic afferent input from that side, creating an acute imbalance between left and right vestibular tone — this does produce nystagmus, but the point being tested is that it is NOT simply 'contralateral nystagmus' in a naive sense; the resulting nystagmus pattern (fast phase typically AWAY from the lesioned side acutely) reflects the loss of input, which is a more nuanced fact than a simple mirrored effect.

PROF'S EMPHASIS: Your professor's cheat-sheet source specifically flags this as a 'does NOT happen' fact — a classic negatively-phrased trap question testing whether you actually understand the tonic-imbalance mechanism instead of just pattern-matching 'vestibular nerve cut = nystagmus.'

Q6. Damage to the vestibular system (semicircular canals, otolith organs, or vestibulocerebellum) can produce pathologic nystagmus. Which...

✓ **Correct: D. Primary auditory cortex**

Pathologic nystagmus specifically traces to damage in the semicircular canals, otolith organs, or vestibulocerebellum — the primary auditory cortex is part of the hearing pathway, not balance, and its damage would not produce this pattern.

Conjugate Eye Movement (5 questions)

Q7. Which cranial nerves are involved in HORIZONTAL conjugate eye movement?

✓ **Correct: A. CN III and CN VI**

Horizontal conjugate eye movement uses CN III (oculomotor, medial rectus for adduction) and CN VI (abducens, lateral rectus for abduction) of the two eyes working together, connected via the MLF.

✗ **B is wrong:** CN III and IV together mediate VERTICAL conjugate movement, not horizontal — a very commonly swapped pair.

PROF'S EMPHASIS: Horizontal = III & VI. Vertical = III & IV. CN IV (trochlear, superior oblique) is the one that's ONLY in the vertical pair, never the horizontal one — a clean way to remember which nerve is the odd one out.

Q8. Which cranial nerves are involved in VERTICAL conjugate eye movement?

✓ **Correct: A. CN III and CN IV**

Vertical conjugate eye movement involves CN III (oculomotor) and CN IV (trochlear, superior oblique) — the pairing tested opposite to the horizontal CN III/VI pair.

✗ **B is wrong:** This is the HORIZONTAL pairing, not vertical.

Q9. What structure carries the signal connecting the cranial nerve nuclei (III, IV, VI) that move the eyeball, coordinating conjugate gaze?

✓ **Correct: A. Medial longitudinal fasciculus (MLF)**

The MLF is the internuclear connection linking the oculomotor, trochlear, and abducens nuclei, allowing them to coordinate conjugate eye movements.

PROF'S EMPHASIS: If a question describes ANY coordinated/conjugate eye movement circuit (horizontal or vertical) linking two of CN III/IV/VI, your first instinct should be 'MLF.'

Q10. Where is the horizontal gaze center (saccadic center) that generates the fast, voluntary horizontal eye movements (saccades)?

✓ **Correct: A. PPRF (Paramedian pontine reticular formation)**

The PPRF (paramedian pontine reticular formation) is the brainstem 'saccade generator' for horizontal conjugate eye movements, working together with the abducens nucleus and MLF.

PROF'S EMPHASIS: PPRF = the saccadic center, explicitly named on your professor's slide. This is a high-yield, specific fill-in-the-blank fact.

Q11. A lesion of the right MLF causes failure of the right eye to adduct on attempted LEFTward conjugate gaze, with nystagmus in the...

✓ **Correct: A. Internuclear ophthalmoplegia (INO)**

This is the textbook definition of internuclear ophthalmoplegia — an MLF lesion disconnects the abducens nucleus (driving the contralateral eye's abduction) from the ipsilateral oculomotor medial rectus subnucleus, so the affected eye fails to adduct on conjugate gaze toward the side opposite the lesion.

Vestibulo-Ocular Reflex (VOR) (5 questions)

Q12. During the horizontal vestibulo-ocular reflex, if the head turns LEFT, what happens to the eyes, and what is the nystagmus called?

✓ **Correct: A. Both eyes turn RIGHT (slow phase, compensatory), and the resulting nystagmus is named 'to the left' (by its fast/reset phase)**

Head turn left → both eyes reflexively turn right (slow, compensatory VOR phase, keeping gaze fixed) → the fast/reset saccadic phase then flicks the eyes back left, and nystagmus is conventionally named by this fast-phase direction — so 'nystagmus to the left.'

***PROF'S EMPHASIS:** This exact head-left/eyes-right/nystagmus-left triad (and its mirror image for head-right) is straight off your professor's slide — she will absolutely test the direction logic here. Draw it out until it's automatic: eyes move OPPOSITE the head (slow phase), nystagmus is named for the FAST phase, which matches the head-turn direction.*

Q13. Which pathway mediates the horizontal VOR? Put it in the correct order.

✓ **Correct: A. Lateral crista ampullares (of the horizontal semicircular canal) → vestibular nuclei in the medulla → oculomotor nucleus & abducens nucleus**

The horizontal VOR circuit: horizontal (lateral) semicircular canal crista ampullaris detects head rotation in the horizontal plane → signal travels to vestibular nuclei in the medulla → which drive the oculomotor and abducens nuclei to produce compensatory horizontal eye movement.

***X D is wrong:** This describes the VERTICAL VOR pathway instead (superior/posterior crista ampullares → vestibular nuclei in the PONS → oculomotor & TROCHLEAR nuclei) — easy to mix up with the horizontal pathway.*

Q14. Which pathway mediates the vertical VOR (head moves up → eyes move down, and vice versa)? Put it in the correct order.

✓ **Correct: A. Superior and posterior crista ampullares → vestibular nuclei in the pons (superior) → oculomotor nucleus & trochlear nucleus**

Vertical VOR: superior and posterior semicircular canal crista ampullares detect vertical-plane head rotation → vestibular nuclei in the pons (superior vestibular nucleus) → drive oculomotor and trochlear nuclei for compensatory vertical eye movement.

***PROF'S EMPHASIS:** Notice the clean split: HORIZONTAL VOR uses the LATERAL canal + medulla + CN III/VI. VERTICAL VOR uses the SUPERIOR/POSTERIOR canals + pons + CN III/IV. Same logic, different canals/level/nerve-pair — learn them as a matched set, not in isolation.*

Q15. A patient's head moves UP. What happens to the eyes via the vertical VOR?

✓ **Correct: A. Eyes move DOWN (compensatory, to keep gaze fixed on the target)**

Vertical VOR is compensatory just like horizontal VOR: head moves up → eyes move down, and head moves down → eyes move up, keeping the visual image stable on the retina despite head movement.

Q16. Which nuclei stimulate the medial and lateral rectus muscles for horizontal VOR, based on the vestibular-nuclei-to-extraocular-muscle...

✓ **Correct: A. Oculomotor nucleus (medial rectus) and abducens nucleus (lateral rectus)**

The oculomotor nucleus drives the medial rectus (adduction) and the abducens nucleus drives the lateral rectus (abduction) — together producing the conjugate horizontal eye movement of the VOR.

Sound-Dampening Mechanism (5 questions)

Q17. What is the physiological purpose of the sound-reduction (“glai lod siang”) mechanism involving the stapedius and tensor tympani reflexes?

✓ **Correct: A. To inhibit hair cell response to certain frequencies, sharpening perception of other, wanted frequencies — e.g. hearing a friend talk in a noisy room**

This 'inhibit sound' circuit selectively dampens certain frequencies via reflex muscle contraction, which relatively sharpens/enhances the frequencies you're trying to attend to — exactly the mechanism your slide describes for picking a voice out of background noise.

Q18. What does contraction of the stapedius muscle (stapedius reflex) mechanically accomplish?

✓ **Correct: A. It holds the stapes firm, preventing the oval window from vibrating too vigorously**

Stapedius contraction pulls on and stiffens the stapes, damping its movement so the oval window isn't driven to vibrate too forcefully by loud sound — a protective/dampening reflex.

✗ **B is wrong:** That's the **TENSOR TYMPANI's** job (via the malleus/tympanic membrane), not the stapedius's (via the stapes/oval window) — a classic swapped-muscle trap.

PROF'S EMPHASIS: Stapedius → stapes → oval window (limits over-vibration there). Tensor tympani → malleus → tympanic membrane tension (adjusts which frequency range comes through well). Keep the muscle-to-structure pairing straight — this is exactly the kind of thing a matching question tests.

Q19. What does contraction of the tensor tympani muscle accomplish?

✓ **Correct: A. It holds the malleus firm, adjusting the tension of the tympanic membrane so it doesn't vibrate too easily, tuning which frequency range is received well**

Tensor tympani contraction stiffens the malleus, changing tympanic membrane tension so the membrane doesn't vibrate too freely — adjusting/tuning the frequency response of the middle ear.

Q20. What is the pathway/circuit driving the sound-reduction (stapedius + tensor tympani) reflex, per your professor's diagram?

✓ **Correct: A. Cochlear nucleus → superior olivary nucleus → reticular formation → CN V (innervating tensor tympani) and CN VII (innervating stapedius)**

The reducing-sound circuit runs cochlear input → superior olivary nucleus → reticular formation → out to CN V (innervating tensor tympani) and CN VII (innervating stapedius) — both cranial nerve motor nuclei are engaged together to dampen sound via their respective middle-ear muscles.

✗ **D is wrong:** This reverses the CN-to-muscle pairing (says CN VII→tensor tympani, CN V→stapedius) — the opposite of standard anatomy. Tensor tympani is trigeminal (V); stapedius is facial (VII). This exact reversal shows up in some hastily-copied student notes, so double check you have it the right way round.

PROF'S EMPHASIS: Watch this one carefully — it's easy to mix up which cranial nerve goes to which muscle. Standard anatomy (and the professor's own labeled diagram) pairs CN V (trigeminal, mandibular branch) with **TENSOR TYMPANI**, and CN VII (facial) with **STAPEDIUS** — the same pairing tested elsewhere (e.g., facial nerve damage causing hyperacusis specifically knocks out the **STAPEDIUS** reflex, because that's the CN VII-innervated one).

Q21. A patient has facial nerve (CN VII) damage and develops hyperacusis (sounds seem abnormally loud). Which specific reflex is lost, and why?

✓ **Correct: A. The stapedius reflex, because CN VII innervates the stapedius muscle**

The stapedius muscle is innervated by CN VII; facial nerve damage disables the stapedius reflex specifically, removing that protective dampening and causing hyperacusis. Tensor tympani (CN V) is unaffected by a CN VII lesion.

✗ **B is wrong:** Tensor tympani is innervated by CN V (trigeminal), not CN VII — a facial nerve lesion doesn't touch it.

Ear Pressure Equalization (3 questions)

Q22. What structure equalizes air pressure between the middle ear cavity and the atmosphere (e.g., the 'ear popping' sensation during...

✓ **Correct: A. Eustachian (auditory/pharyngotympanic) tube, connecting the middle ear to the nasopharynx**

The Eustachian tube connects the middle ear cavity to the nasopharynx, allowing air pressure to equalize with the atmosphere — its dysfunction (e.g., from congestion) is a common cause of middle-ear pressure symptoms and conductive hearing changes.

***PROF'S EMPHASIS:** If a question mentions altitude change, diving, or 'ears popping,' the answer is the Eustachian tube — essentially every time.*

Q23. Blockage of the Eustachian tube (e.g., from an upper respiratory infection) would be expected to cause which type of hearing change?

✓ **Correct: A. Conductive hearing changes, from altered middle-ear pressure and reduced tympanic membrane mobility**

Eustachian tube blockage traps air in the middle ear at an abnormal pressure, restricting the eardrum/ossicular chain's normal mobility — a conductive (outer/middle ear) mechanism, not a sensorineural (cochlear/nerve) one.

Q24. Which best distinguishes the round window's function from the oval window's, in the context of pressure within the cochlea?

✓ **Correct: A. The oval window is where the stapes pushes fluid IN; the round window bulges OUT to release that pressure wave, allowing continued fluid movement**

The stapes footplate pushes fluid inward at the oval window; because cochlear fluid is incompressible, the round window membrane must bulge outward to accommodate that pressure wave, allowing continuous fluid movement through the cochlea.

***PROF'S EMPHASIS:** Oval = IN (stapes pushes). Round = OUT (releases the wave). This is a completely different pressure system from the Eustachian tube (air pressure) — don't confuse the two 'pressure' questions.*

Cochlear Base/Apex Frequency Tuning (4 questions)

Q25. Which part of the cochlea responds best to HIGH-frequency sound, and why?

✓ **Correct: A. The base — the basilar membrane there is short and taut (stiff)**

At the cochlear base, the basilar membrane fibers are short and under more tension (taut) — this mechanical stiffness makes them resonate best to high-frequency vibration, exactly as your professor's slide states.

***PROF'S EMPHASIS:** 'Short and tight/tense strings ring high; long and loose strings ring low' — same physical principle as a guitar string. Base = short & taut = high frequency. Apex = long & floppy = low frequency.*

Q26. Which part of the cochlea responds best to LOW-frequency sound, and why?

✓ **Correct: A. The apex — the basilar membrane fibers there are long and loose (floppy)**

At the cochlear apex, basilar membrane fibers are long and loose (less tension), which resonate best to low-frequency vibration — the opposite mechanical setup from the base.

Q27. Why is knowing the cochlea's tonotopic (base=high, apex=low) organization clinically useful when a patient has a middle-ear injury...

✓ **Correct: A. The pattern of which frequencies are lost on a hearing test can localize WHERE along the cochlea the damage occurred**

Because specific cochlear locations are tuned to specific frequencies, an audiogram showing loss at particular frequencies can help localize exactly where along the cochlea (base vs. apex vs. middle) the injury or degeneration has occurred — a direct clinical application of tonotopy.

Q28. A patient develops high-frequency hearing loss first (as in presbycusis). Based on cochlear tonotopy, which region is most affected?

✓ **Correct: A. The base of the cochlea**

High-frequency loss localizes to the cochlear base, which is tuned to high frequencies — matching presbycusis's classic pattern of cumulative base-first hair cell damage.

Head Rotation — Which Structure Responds (5 questions)

Q29. Rotating the head horizontally (e.g., shaking the head 'no') primarily stimulates which structure?

✓ **Correct: A. The lateral (horizontal) semicircular canal**

Horizontal head rotation is detected by the lateral (horizontal) semicircular canal, whose crista ampullaris senses angular acceleration in that plane — this is also the canal driving the horizontal VOR.

***PROF'S EMPHASIS:** Per your professor's own cheat-sheet phrasing: 'Head rotation → Lateral semicircular canal' — if a question just says 'head rotation' with no further plane specified, this is the default expected answer.*

Q30. Flexing the head to touch the chin toward the shoulder (a diagonal/vertical-plane rotational movement) stimulates which structure, per...

✓ **Correct: A. Posterior semicircular canal**

This specific movement example (flexing the head toward the shoulder) is used in your lecture to illustrate posterior semicircular canal stimulation — a vertical/diagonal-plane rotational movement.

Q31. During which condition do the semicircular canals NOT generate extra afferent nerve impulses above baseline?

✓ **Correct: A. Constant-velocity rotation (steady-state spinning), because the canals only detect DYNAMIC changes in velocity — acceleration and deceleration — not constant velocity itself**

Semicircular canal hair cells respond to the cupula being deflected by relative endolymph flow, which only happens during changes in rotational velocity (acceleration/deceleration) — once rotation becomes constant, the endolymph 'catches up' to the canal wall, the cupula returns to a neutral position, and afferent firing returns to baseline despite ongoing rotation.

***PROF'S EMPHASIS:** This is an explicitly flagged, favorite trick question from your professor: canals are ACCELERATION detectors, not velocity detectors. Constant velocity = no net cupula deflection = afferent firing returns to baseline, even though the head is still very much rotating.*

Q32. Which vestibular nucleus receives input from the posterior semicircular canal AND the saccule?

✓ **Correct: A. Inferior vestibular nucleus**

Per your lecture material, the inferior vestibular nucleus is the synaptic target for both the posterior semicircular canal and the saccule.

✗ B is wrong: The superior vestibular nucleus instead receives from the superior/lateral semicircular canals and the utricle — the complementary pairing.

***PROF'S EMPHASIS:** Pair these two facts as a set: Inferior vestibular nucleus = posterior canal + saccule. Superior vestibular nucleus = superior/lateral canals + utricle. This exact pairing is specifically called out in your source material.*

Q33. Which vestibular nucleus receives input from the superior/lateral semicircular canals AND the utricle?

✓ **Correct: A. Superior vestibular nucleus**

Per your lecture material, the superior vestibular nucleus receives input from the superior and lateral semicircular canals plus the utricle — the complementary pairing to the inferior vestibular nucleus (posterior canal + saccule).

Elevator / Driving — Linear Acceleration (4 questions)

Q34. Riding in an elevator that suddenly accelerates upward primarily stimulates which structure?

✓ **Correct: A. The saccule (vertical linear acceleration)**

The saccule's macula is oriented in a vertical plane, making it the primary detector of vertical linear acceleration — exactly what an elevator starting or stopping produces.

✗ **B is wrong:** The utricle detects **HORIZONTAL** linear acceleration (like a car pulling away from a stoplight) — the perpendicular plane from what an elevator produces.

PROF'S EMPHASIS: Saccule = Straight up-and-down (vertical linear, elevator). Utricle = yoU're driving forward (horizontal linear, car). This pairing is tested from multiple angles — know it cold.

Q35. A car accelerating forward from a stop primarily stimulates which structure?

✓ **Correct: A. The utricle (horizontal linear acceleration)**

The utricle's macula lies in the horizontal plane, making it the primary sensor for horizontal linear acceleration — exactly what happens when a car speeds up or slows down.

✗ **D is wrong:** The lateral semicircular canal detects **ANGULAR** (rotational) acceleration, like turning the head or the car spinning — not straight-line forward acceleration.

Q36. What are the otolith organs (utricle and saccule) structurally specialized with, that gives them the inertial mass needed to detect...

✓ **Correct: A. Otoconia (calcium carbonate crystals) embedded in an otolithic membrane overlying the hair cell stereocilia**

Otoconia (calcium carbonate crystals) embedded in the otolithic membrane give the maculae extra inertial mass, so the membrane lags behind during linear acceleration/gravity shifts, deflecting the underlying stereocilia — the ampullary cupula of the semicircular canals, by contrast, has no such crystals.

Q37. Walking forward at a steady pace (repeated small forward accelerations/decelerations with each step) is most associated with stimulation...

✓ **Correct: A. Utricle (macula)**

Walking forward involves horizontal linear acceleration/deceleration with each stride, detected by the utricle, consistent with the same horizontal-linear-acceleration logic as a car pulling away.

PROF'S EMPHASIS: Anchor this with the same rule as 'driving a car' — forward/horizontal linear acceleration always points to the utricle.

PART 2 — Eye

Eye Tissue Layers: Light Absorption & Phototransduction (7 questions)

Q38. The eye is composed of three tissue layers (tunics). Which layer is the middle layer, and what does it consist of?

✓ **Correct: A. Tunica vasculosa (uvea) — choroid, ciliary body, and iris**

The eye's three tunics are: outer tunica fibrosa (sclera + cornea), middle tunica vasculosa/uvea (choroid + ciliary body + iris), and the innermost retina (pigmented layer + neural layer).

PROF'S EMPHASIS: Three layers, outside-in: Tunica FIBROSA (sclera+cornea, tough outer coat) → Tunica VASCULOSA/uvea (choroid+ciliary body+iris, the vascular middle layer) → Retina (pigmented + neural layers, innermost).

Q39. Which layer of the eye is primarily responsible for ABSORBING stray light, preventing internal scatter/reflection that would blur the image?

✓ **Correct: A. Choroid (heavily pigmented, part of the tunica vasculosa)**

The heavily pigmented choroid absorbs light that has passed through the retina, preventing it from scattering/reflecting back and degrading image quality.

PROF'S EMPHASIS: Same logic as a camera's matte-black interior — the pigmented choroid stops light from bouncing around inside the eye after it's already been sensed.

Q40. Which structure actually converts light energy into a nerve impulse (phototransduction)?

✓ **Correct: A. Photoreceptor cells (rods and cones) in the retina, via their photopigments**

Rods and cones contain photopigments (e.g., rhodopsin) that undergo a light-induced structural change, triggering the phototransduction cascade that ultimately produces a change in membrane potential (a graded receptor potential) — the actual light-to-electrical conversion step.

Q41. Put the phototransduction cascade events in the CORRECT order, from light hitting the photopigment to the final electrical change.

✓ **Correct: A. Structural change of the photopigment → activation of transducin → activation of phosphodiesterase → decreased intracellular cGMP → closure of Na⁺ channel**

This is the canonical phototransduction cascade: light causes a structural change in the photopigment (e.g., rhodopsin bleaching) → this activates the G-protein transducin → transducin activates phosphodiesterase (PDE) → PDE breaks down cGMP, decreasing its intracellular concentration → falling cGMP closes cGMP-gated Na⁺ channels → the photoreceptor hyperpolarizes. Note this is the OPPOSITE of typical sensory transduction — light causes HYPERPOLARIZATION, not depolarization.

PROF'S EMPHASIS: This EXACT ordering question has shown up before in your archive (professor loves it) with the note 'answer: 4>1>3>5>2' referring to this same 5-step list — memorize the full chain, not just pieces: Photopigment changes shape → activates transducin (a G-protein) → which activates phosphodiesterase → which breaks down cGMP (decreasing it) → closing cGMP-gated Na⁺ channels (hyperpolarizing the photoreceptor).

Q42. Unlike most sensory receptors, photoreceptors respond to their stimulus (light) with which membrane change?

✓ **Correct: A. Hyperpolarization (light closes Na⁺ channels, reducing the dark current)**

In darkness, cGMP-gated Na⁺ channels are open, producing a depolarizing 'dark current'; light triggers the cascade that closes these channels, so light causes hyperpolarization, not depolarization — the opposite of most other sensory transduction.

PROF'S EMPHASIS: This is one of the most commonly tested 'gotcha' facts in visual physiology: photoreceptors are DEPOLARIZED in the dark (constant 'dark current' from open cGMP-gated channels) and HYPERPOLARIZE in response to light, the reverse of almost every other sensory receptor type you've learned.

Q43. Which retinal cell layer contains the cell bodies of the photoreceptors themselves?

✓ **Correct: A. Outer nuclear layer**

The outer nuclear layer contains the cell bodies of rods and cones; the inner nuclear layer contains bipolar, horizontal, and amacrine cell bodies — a commonly confused pair.

✗ **B is wrong:** The inner nuclear layer holds bipolar/horizontal/amacrine cell bodies, not photoreceptor cell bodies.

Q44. Correctly order the retinal layers from the rod/cone layer toward the optic nerve fiber layer.

✓ **Correct: A. Rods/cones → outer nuclear layer → inner nuclear layer → ganglion cell layer → optic nerve fiber layer**

Correct order from photoreceptor segments toward the vitreous: rods/cones → outer nuclear layer (photoreceptor bodies) → inner nuclear layer (bipolar/horizontal/amacrine bodies) → ganglion cell layer → optic nerve fiber layer.

Visual Pathway Lesions — Specific Terminology (10 questions)

Q45. A lesion of the OPTIC NERVE (before the chiasm) causes which type of visual field defect?

✓ **Correct: A. Monocular blindness (or a monocular field defect) in the eye on the same side as the lesion**

The optic nerve carries information from only ONE eye (organized 'by eye,' not yet reorganized by visual field), so damage there produces a monocular deficit — vision loss confined to that one eye.

PROF'S EMPHASIS: Sort every visual pathway question into ONE of three zones first: (1) PRE-chiasm (optic nerve) = monocular. (2) AT the chiasm = heteronymous (bitemporal or binasal). (3) POST-chiasm (tract/radiation/cortex) = homonymous. Everything else follows from this.

Q46. A lesion at the MIDLINE of the optic chiasm (e.g., from a pituitary tumor pressing upward) classically causes which visual field defect?

✓ **Correct: A. Bitemporal heteronymous hemianopia**

At the midline of the chiasm, the crossing fibers are the ones from the NASAL retina of each eye (carrying the TEMPORAL visual field) — a midline lesion damages these crossing fibers bilaterally, producing loss of both temporal visual fields (bitemporal hemianopia). This is the classic pituitary adenoma presentation.

✗ **B is wrong:** *Binasal hemianopia results from a LATERAL (not midline) chiasm lesion compressing the uncrossed temporal-retina fibers on both sides — a real but much less commonly tested pattern, easy to confuse with the midline/bitemporal case.*

PROF'S EMPHASIS: Midline chiasm compression (pituitary tumor, growing UP into the chiasm) = bitemporal. Lateral chiasm compression (e.g., bilateral carotid artery aneurysms pressing in from the SIDES) = binasal, the much rarer opposite-pattern lesion your professor also tests — don't forget this second, less common pairing exists.

Q47. A lesion at the LATERAL part of the optic chiasm (e.g., bilateral compressive lesions from the sides) classically causes which visual...

✓ **Correct: A. Binasal heteronymous hemianopia**

The lateral chiasm carries the UNCROSSED fibers (from the temporal retina, carrying the NASAL visual field) that don't cross at the chiasm at all — bilateral lateral compression damages these, causing loss of both nasal visual fields (binasal hemianopia), the mirror-image pattern to the far more common bitemporal presentation.

Q48. A lesion of the OPTIC TRACT (just after the chiasm) causes which visual field defect?

✓ **Correct: A. Homonymous hemianopia (complete half-field loss in both eyes, on the side opposite the lesion), WITHOUT macular sparing**

Post-chiasmal structures carry a mixed, binocular representation of the opposite visual field, so optic tract damage causes a homonymous (matching, same-side-in-both-eyes) field cut. Because the tract is a single compact bundle (not yet given the dual blood supply protection that the occipital cortex has), this loss is typically COMPLETE, without the macular sparing seen in cortical lesions.

✗ **E is wrong:** *Macular sparing is specifically a feature of OCCIPITAL CORTEX lesions (dual blood supply), not optic tract lesions — mixing these two up is exactly the kind of distinction your professor is testing.*

PROF'S EMPHASIS: Optic TRACT lesion = homonymous hemianopia, NO macular sparing (the tract carries a compact, unified bundle including macular fibers, so it's usually all-or-nothing). This is the key contrast your professor wants you to know against the CORTICAL (occipital) lesion pattern, which DOES spare the macula.

Q49. A lesion of the OPTIC RADIATION (complete, affecting both the parietal and temporal fibers) causes which visual field defect?

✓ **Correct: A. Homonymous hemianopia (on the side opposite the lesion)**

A complete lesion of the optic radiation (affecting all its fibers, both the parietal 'inferior field' fibers and the temporal/Meyer's loop 'superior field' fibers) still produces a full homonymous hemianopia, just as a tract lesion would — quadrantanopia only results from a PARTIAL/selective radiation lesion.

PROF'S EMPHASIS: Note the pattern: complete optic tract damage OR complete optic radiation damage both give homonymous HEMIAnopia. Only a PARTIAL/selective radiation lesion (hitting just Meyer's loop OR just the parietal fibers) gives a QUADRANTAnopia instead.

Q50. A SELECTIVE lesion of the optic radiation fibers (i.e., only part of it, such as Meyer's loop alone) causes which visual field defect?

✓ **Correct: A. A quadrantanopia (loss of one specific quadrant of the visual field)**

Because the optic radiation fibers spread out and separate by visual field quadrant as they travel (temporal lobe fibers carry the superior field via Meyer's loop, parietal fibers carry the inferior field), damaging only PART of the radiation knocks out only the corresponding QUADRANT, not the whole half-field.

Q51. Which structure, if damaged, produces a homonymous hemianopia WITH macular sparing — the pattern your professor specifically emphasizes?

✓ **Correct: A. The occipital (visual) cortex — because the macular region has a dual blood supply (both MCA and PCA collateral flow), it's often spared even when the rest of the visual cortex is infarcted**

The occipital pole (representing central/macular vision) receives blood supply from both the posterior cerebral artery (main occipital supply) AND collateral flow from the middle cerebral artery — so a PCA stroke affecting the rest of the occipital cortex often spares this dual-supplied macular region, producing a homonymous hemianopia that spares central vision. This is the single most clinically useful way to localize a homonymous hemianopia to the CORTEX specifically, rather than to the optic tract.

✗ *D is wrong: An optic tract lesion causes homonymous hemianopia too, but classically WITHOUT macular sparing — the opposite finding from what a cortical lesion produces.*

PROF'S EMPHASIS: IMPORTANT — THIS IS THE ONE YOUR PROFESSOR SAID SHE EMPHASIZES. Macula sparing = occipital cortex lesion (dual blood supply protects the macular representation). NO macula sparing = optic tract lesion (single, unprotected bundle). If a question gives you 'homonymous hemianopia' and doesn't mention macular sparing either way, check carefully whether the vignette is describing a cortical stroke (think PCA occlusion) vs. a tract lesion — that's exactly the distinction being tested.

Q52. A patient has a RIGHT occipital lobe infarct (PCA territory). What visual field defect is expected?

✓ **Correct: A. Left homonymous hemianopia WITH macular sparing**

A right occipital lesion causes a LEFT-sided (contralateral) homonymous field defect, and because it's a cortical (not tract) lesion, the dual blood supply to the macular region typically spares central vision — left homonymous hemianopia with macular sparing.

✗ *C is wrong: This gets the side right but wrongly denies macular sparing — occipital cortex lesions ARE the classic macula-sparing lesion, unlike optic tract lesions.*

Q53. A lesion of the RIGHT Meyer's loop (temporal lobe portion of the optic radiation) causes which specific visual field defect?

✓ **Correct: A. Left homonymous SUPERIOR quadrantanopia (pie in the sky)**

Meyer's loop carries fibers for the CONTRALATERAL SUPERIOR visual field (looping through the temporal lobe); a right-sided lesion therefore produces a LEFT, SUPERIOR quadrant defect.

PROF'S EMPHASIS: Temporal lobe (Meyer's loop) = 'pie in the SKY' = contralateral SUPERIOR quadrant. Parietal lobe fibers = 'pie on the FLOOR' = contralateral INFERIOR quadrant. This exact question has appeared repeatedly in your archive — know both halves of this pairing.

Q54. A lesion of the PARIETAL lobe fibers of the optic radiation (superior fibers, NOT Meyer's loop) causes which visual field defect?

✓ **Correct: A. A contralateral INFERIOR quadrantanopia (pie on the floor)**

Parietal lobe optic radiation fibers carry the contralateral INFERIOR visual field — damage here gives the mirror-image quadrant defect to a Meyer's loop lesion.

Meyer's Loop & Macula Sparing — Prof's Emphasis (5 questions)

Q55. Why does the OPTIC TRACT lesion NOT spare the macula, while an OCCIPITAL CORTEX lesion classically DOES?

✓ **Correct: A. The optic tract is a single, compact bundle with a single blood supply and no built-in redundancy for any part of the visual field, including the macula; the occipital cortex's macular representation specifically benefits from dual (MCA + PCA) blood supply not available further upstream**

Macular sparing is fundamentally a VASCULAR phenomenon, not a purely anatomical one: the occipital pole (where the macula is represented) gets blood from both the PCA (main supply) and collateral MCA branches, so a PCA stroke often spares it. The optic tract, chiasm, and most of the radiation don't have this same dual-supply protection, so ischemic or compressive lesions there tend to take out the macular fibers along with everything else.

PROF'S EMPHASIS: This is a vascular anatomy explanation layered on top of a fiber anatomy fact — exactly the kind of 'why' integration your professor rewards. It's not that macular fibers are special anatomically until you reach the cortex; it's that the CORTEX specifically has a back-up blood supply that upstream structures (chiasm, tract, radiation) don't have.

Q56. A 65-year-old patient has a stroke and is found to have a complete left homonymous hemianopia, but central (macular) vision is preserved...

✓ **Correct: A. Right occipital cortex (likely right PCA territory)**

Macular sparing localizes the lesion to the occipital CORTEX rather than the optic tract (a tract lesion would NOT spare the macula); the LEFT-sided field loss means the lesion is on the RIGHT side (post-chiasmal structures serve the contralateral field).

✗ **B is wrong:** An optic tract lesion could also give left homonymous hemianopia (if right-sided), but would NOT be expected to spare the macula — the macular sparing detail is exactly what rules this out and points to the cortex instead.

Q57. A lesion of the right optic tract (as opposed to the right occipital cortex) causes left homonymous hemianopia. What KEY difference...

✓ **Correct: A. The optic tract lesion does NOT spare the macula, while the occipital cortex lesion classically DOES**

Both lesions can produce a left homonymous hemianopia, but the presence or absence of macular sparing is the key bedside clue distinguishing WHERE along the pathway the lesion sits — tract (no sparing) vs. cortex (sparing, due to dual blood supply).

Q58. Which best describes why 'macula sparing' is clinically valuable, beyond just being a testable fact?

✓ **Correct: A. It helps localize a homonymous hemianopia to the cortex (implying a vascular/PCA stroke mechanism) rather than to the optic tract (implying a different, often compressive or different-vascular-territory process), guiding further workup**

Localizing a homonymous hemianopia to the cortex versus the tract has real diagnostic implications (different likely causes, different vascular territories, different associated findings) — this is why your professor emphasizes it, not just as an isolated fact but as a genuine localizing tool.

Q59. Combining what you know: a patient has a LEFT homonymous INFERIOR quadrantanopia. Which structure is most likely damaged, and on which side?

✓ **Correct: A. Right parietal lobe optic radiation fibers (carrying the contralateral inferior field)**

Inferior quadrant defects come from PARIETAL lobe optic radiation fibers (not Meyer's loop, which gives superior defects); since it's a homonymous (post-chiasmal) defect on the LEFT, the lesion must be on the RIGHT side.

PROF'S EMPHASIS: Work every quadrantanopia question the same way: (1) which quadrant is affected? (2) is it superior (temporal/Meyer's) or inferior (parietal)? (3) which SIDE (opposite the lesion, since it's homonymous/post-chiasmal)? Three-step logic, every time, rather than memorizing each combination as a separate fact.

PART 3 — Ear Development

Tri-Germ-Layer Structures (Tympanic Membrane) (4 questions)

Q60. The tympanic membrane (eardrum) is a rare structure that develops from all THREE germ layers. Match each layer to the tympanic membrane...

✓ **Correct: A. Outer layer = ectoderm (from the 1st pharyngeal cleft); middle layer = mesoderm (fibrous connective tissue); inner layer = endoderm (from the 1st pharyngeal pouch)**

The tympanic membrane sits at the junction between the ectoderm-derived external ear (1st pharyngeal cleft) and the endoderm-derived middle ear cavity (1st pharyngeal pouch) — its outer epithelial layer is ectoderm, its inner epithelial layer is endoderm, and a middle fibrous connective tissue layer between them is mesoderm-derived. This tri-laminar, tri-germ-layer structure directly reflects its position at that developmental boundary.

PROF'S EMPHASIS: The tympanic membrane is *THE* classic 'tri-germ-layer' structure in ear embryology — exactly the kind of fact a professor loves because it forces you to remember all three layers *AND* their spatial order (outer=ectoderm, matching the external ear's cleft origin; inner=endoderm, matching the middle ear's pouch origin; middle=mesoderm, the connective tissue sandwiched between).

Q61. Which structure marks the boundary between the ectoderm-derived external ear and the endoderm-derived middle ear during development?

✓ **Correct: A. The tympanic membrane**

The tympanic membrane physically and developmentally sits at the junction of the first pharyngeal cleft (external ear, ectoderm) and the first pharyngeal pouch (middle ear, endoderm) — exactly why it uniquely draws on both germ layers, plus mesoderm for its fibrous middle layer.

Q62. Which ear structures develop from the first pharyngeal CLEFT (ectoderm)?

✓ **Correct: A. The external auditory meatus (and contributes to the outer layer of the tympanic membrane)**

The first pharyngeal cleft, lined by ectoderm, forms the external auditory meatus and contributes the outer epithelial layer of the tympanic membrane.

✗ **C is wrong:** Tympanic cavity and Eustachian tube come from the first pharyngeal *POUCH* (endoderm), not the cleft — the paired, opposite structure.

PROF'S EMPHASIS: Cleft = ectoderm = *OUTER* ear structures (external auditory meatus, outer TM layer). Pouch = endoderm = *MIDDLE* ear structures (tympanic cavity, Eustachian tube, inner TM layer). Don't mix up cleft vs. pouch — it's a classic swap.

Q63. Which ear structures develop from the first pharyngeal POUCH (endoderm)?

✓ **Correct: A. The tympanic cavity (middle ear space) and auditory (Eustachian) tube, plus the inner epithelial layer of the tympanic membrane**

The first pharyngeal pouch, lined by endoderm, expands to form the tympanic cavity and auditory tube, and contributes the inner epithelial lining of the tympanic membrane — the endodermal counterpart to the ectodermal external ear.

Ectoderm / Mesoderm / Endoderm — General Derivatives (6 questions)

Q64. In general terms, what does surface ECTODERM give rise to, among structures relevant to the special senses?

✓ **Correct: A. Lens of the eye, corneal epithelium, external ear/auditory meatus, olfactory epithelium, and (with endoderm/mesoderm) the outer tympanic membrane layer**

Surface ectoderm forms epithelial surface structures like the lens (via lens placode), corneal epithelium, external auditory meatus lining, and olfactory epithelium — distinct from *NEUROectoderm*, which forms the retina/optic nerve and the otic vesicle/membranous labyrinth.

✗ **B is wrong:** Choroid, sclera, and corneal endothelium are *NEURAL CREST* derivatives (migrating mesenchyme), not surface ectoderm.

PROF'S EMPHASIS: Surface ectoderm = the 'outer-facing epithelial' contributions: lens, corneal epithelium, external ear canal, olfactory epithelium, nasal placodes, outer TM layer. Keep this separate from *NEUROectoderm* (retina, optic nerve, otic vesicle/membranous labyrinth) — same germ layer family, different sub-category, very easy to blur together.

Q65. What does *NEUROectoderm* give rise to, among special-sense structures?

✓ **Correct: A. The retina, optic nerve, and the otic vesicle/membranous labyrinth (inner ear) — both being direct outgrowths of neural tissue**

The optic vesicle/cup (retina, optic nerve) is literally an outgrowth of the diencephalon (neuroectoderm); the otic vesicle/otocyst (forming the entire membranous labyrinth of the inner ear) arises from the otic placode, which invaginates from surface ectoderm but is functionally/developmentally grouped with neural tissue given its origin and behavior.

PROF'S EMPHASIS: This is why the inner ear (via the otic PLACODE, a specialized patch of surface ectoderm that invaginates and behaves like neuroectoderm) and the eye's retina (a direct diencephalic outgrowth) are both considered 'part of the CNS' in a real embryological sense — worth remembering when a question asks 'which special sense organ is most directly derived from the brain itself.'

Q66. What does MESODERM contribute to the special sense organs?

✓ **Correct: A. Connective tissue and vascular contributions — e.g., the fibrous/vascular coats of the eye, the middle layer of the tympanic membrane, and (with neural crest) much of the periocular/perilabyrinthine mesenchyme**

Mesoderm (often working alongside neural crest-derived mesenchyme) forms connective tissue, vascular, and structural support elements — the fibrous/vascular coats of the eye, the tympanic membrane's middle fibrous layer, and similar supportive tissue around developing sense organs.

Q67. What does ENDODERM contribute to the ear, specifically?

✓ **Correct: A. The middle ear cavity (tympanic cavity) and auditory tube, plus the inner layer of the tympanic membrane, all via the first pharyngeal pouch**

Endoderm's contribution to the ear comes specifically through the first pharyngeal pouch, forming the tympanic cavity, auditory (Eustachian) tube, and the inner epithelial layer of the tympanic membrane.

Q68. Which eye structures are neural CREST derivatives (as opposed to surface ectoderm, neuroectoderm, or mesoderm proper)?

✓ **Correct: A. Corneal endothelium, choroid, sclera, and much of the corneal stroma**

Migrating neural crest cells differentiate into the corneal endothelium, choroid, sclera, and much of the corneal stroma — connective tissue contributions that are commonly (and easily) mistaken for pure mesoderm.

PROF'S EMPHASIS: Neural crest is often taught as a 'fourth germ layer' for exactly this reason — it contributes extensively to eye connective tissue (corneal endothelium, choroid, sclera) that a simple three-germ-layer framework would otherwise misattribute to mesoderm alone.

Q69. True or false framing: since the tympanic membrane draws on all three germ layers, which single statement correctly captures why this is...

✓ **Correct: A. Most individual adult structures are derived from predominantly ONE germ layer; the tympanic membrane is a rare example of a single structure requiring coordinated contribution from all three (ectoderm, mesoderm, endoderm), reflecting its position exactly at the ectoderm/endoderm boundary between external and middle ear**

Most mature structures trace predominantly to a single germ layer (or germ layer + neural crest); the tympanic membrane's requirement for all three primary germ layers is what makes it a favorite 'integrative embryology' teaching example — a direct physical record of it forming exactly at the ecto/endoderm boundary between the pharyngeal cleft and pouch.

PART 4 — Somatosense

Protopathic vs. Epicritic Sensation (5 questions)

Q70. Which sensations are classified as PROTOPATHIC (affective) sensation, per your professor's table?

✓ **Correct: A. Pain, hot & cold (temperature extremes), crude/gross touch**

Per your lecture table: protopathic (affective) sensations = pain, hot & cold, and crude/gross touch — primitive sensations without fine detail.

PROF'S EMPHASIS: Protopathic = primitive/no-detail sense: pain, hot/cold, crude touch. Epicritic = discriminative/detailed sense: fine touch, pressure, vibration, two-point discrimination, proprioception. Your professor's exact table splits it this way — memorize the table, not a vague 'pain vs touch' shortcut.

Q71. Which sensations are classified as DISCRIMINATIVE (epicritic) sensation?

✓ **Correct: A. Fine/light touch, pressure, vibration, two-point discrimination, and proprioceptive sense**

Epicritic (discriminative) sensations = fine/light touch, pressure, vibration, two-point discrimination, and proprioception — the more detailed, evolutionarily newer sensory modalities.

Q72. Structurally, how do protopathic and epicritic receptors/fibers differ?

✓ **Correct: A. Protopathic: no receptor covering, nonmyelinated axon, branching nerve endings. Epicritic: receptor has a covering/capsule, myelinated axon, NO branching of the nerve ending**

Protopathic receptors have no covering, use nonmyelinated axons, and show branching nerve endings at the receptor (e.g., free nerve endings) — matching their poorly-localized, primitive character. Epicritic receptors have a covering/capsule, myelinated axons, and NO branching at the receptor (e.g., a single encapsulated ending) — matching their precise, well-localized character.

PROF'S EMPHASIS: Free nerve endings (protopathic) = no capsule + branching + unmyelinated. Encapsulated receptors (epicritic, e.g. Meissner's, Pacinian) = capsule + no branching + myelinated. This structure-function link is exactly why protopathic sensation is 'primitive/poorly localized' and epicritic is 'precise' — the anatomy explains the function.

Q73. Within the dorsal root of a spinal nerve, epicritic and protopathic fibers enter via different divisions. Which is correct?

✓ **Correct: A. Epicritic fibers (large, myelinated) enter the MEDIAL division of the posterior rootlet; protopathic fibers (small, nonmyelinated) enter the LATERAL division**

Large myelinated (epicritic) fibers enter medially in the posterior rootlet; small nonmyelinated (protopathic) fibers enter laterally — this segregation at the root sets up their eventual separation into different ascending pathways (dorsal columns vs. spinothalamic tract).

PROF'S EMPHASIS: Medial = big & myelinated = epicritic. Lateral = small & unmyelinated = protopathic. This medial/lateral split at root entry is also WHY the two sensation types can travel via different central pathways (DCML vs. spinothalamic) once inside the cord — it's the anatomical starting point for that whole downstream separation.

Q74. Within the dorsal root/spinal cord entry zone, sensory fiber types are also organized medial-to-lateral by MODALITY. Which is the...

✓ **Correct: A. Unconscious proprioception → conscious proprioception → touch/pressure → pain & temperature**

Per your lecture's diagram, the medial-to-lateral order at the dorsal root entry zone is: unconscious proprioception, then conscious proprioception, then touch/pressure, then pain & temperature (most lateral) — consistent with pain/temperature being protopathic (lateral-entering) fibers.

PROF'S EMPHASIS: Medial-to-lateral: (unconscious) proprioception → (conscious) proprioception → touch/pressure → pain/temperature. Pain and temperature fibers are the MOST LATERAL of the group — consistent with protopathic fibers entering the lateral division discussed above.

The Four Types of Pain (6 questions)

Q75. Your professor's lecture describes FOUR types of pain. Which of the following correctly lists all four?

✓ **Correct: A. Superficial pain (with fast and slow subtypes), Deep pain, Visceral pain, and Referred pain**

Your lecture specifically teaches four pain categories: superficial pain (itself divided into fast and slow subtypes), deep pain, visceral pain, and referred pain — each with distinct mechanisms and clinical features.

PROF'S EMPHASIS: Four types: (1) Superficial [ectoderm-derived skin, itself split into fast/Ad and slow/C], (2) Deep [mesoderm-derived muscle/tendon/joint], (3) Visceral, (4) Referred. This exact four-part scheme is your professor's own classification — learn all four names, not just fast/slow.

Q76. SUPERFICIAL pain arises from which germ layer, and what are its two subtypes?

✓ **Correct: A. Ectoderm; fast pain (myelinated A-delta fibers, sharp/well-localized) and slow pain (unmyelinated C fibers, dull/poorly localized)**

Superficial pain arises from ectoderm-derived skin, with fast pain (sharp, well-localized, carried by myelinated A-delta fibers) and slow pain (dull, aching, poorly-localized, carried by unmyelinated C fibers) as its two subtypes — chemically mediated in part by histamine.

PROF'S EMPHASIS: Superficial pain = skin = ECTODERM. Deep pain = muscle/tendon/joint = MESODERM. This germ-layer link is a distinctive feature of how your professor frames pain — not all courses teach it this way, so it's worth specifically remembering FOR THIS EXAM.

Q77. DEEP pain arises from which germ layer and which structures, and what is its characteristic clinical presentation?

✓ **Correct: A. Mesoderm; from muscle, tendon, joint, periosteum, and ligament (e.g., ischemia/muscle cramp); poorly localized, deep/dull ache over a wide area, often WITH autonomic symptoms (nausea, sweating, low BP, slow HR)**

Deep pain arises from mesoderm-derived structures (muscle, tendon, joint, periosteum, ligament) — classic examples include ischemic muscle cramps. It's characteristically deep, dull, and poorly localized over a wide area, and is often accompanied by autonomic nervous system symptoms: nausea, sweating, low blood pressure, and slowed heart rate.

PROF'S EMPHASIS: Deep pain's autonomic symptom package (nausea, sweating, hypotension, bradycardia) is a specific, testable detail — don't just remember 'poorly localized,' remember the ANS symptoms too, since that's exactly the kind of specific detail an exam question will hinge on.

Q78. VISCERAL pain has which characteristic features, per your lecture?

✓ **Correct: A. Triggered by mechanical distension or ischemia; few receptors so poorly localized; conducted via CN VII, IX, X and the autonomic nervous system to the thalamus and then postcentral gyrus (areas 3,1,2)**

Visceral pain is triggered by mechanical distension or ischemia of internal organs; because visceral tissue has relatively few pain receptors, it's poorly localized; the signal travels via cranial nerves VII, IX, and X along with the autonomic nervous system, ultimately reaching the thalamus and then the postcentral gyrus (Brodmann areas 3, 1, 2).

Q79. What is REFERRED pain, and what theory explains it?

✓ **Correct: A. Somatic pain felt in a body region that shares the same dermatome as an internal organ experiencing visceral or deep pain, explained by convergence theory and facilitation theory**

Referred pain occurs when visceral or deep pain from an organ is perceived as somatic pain in a different body region sharing the same dermatome/spinal segment — explained by convergence theory (visceral and somatic afferents converging onto the same second-order neuron, so the brain 'misattributes' the source) and facilitation theory. Your lecture's specific example: a decayed tooth can cause referred pain felt at the ear pinna, via convergence in the spinal trigeminal nucleus and tract.

PROF'S EMPHASIS: Classic example straight from your slide: a decayed tooth causing pain felt at the ear pinna, via convergence within the spinal trigeminal nucleus/tract. If a vignette describes pain felt somewhere that doesn't match the actual site of pathology, think 'referred pain' + convergence theory immediately.

Q80. Which theory of referred pain proposes that visceral and somatic sensory afferents converge onto the SAME second-order (or higher)...

✓ **Correct: A. Convergence theory**

Convergence theory explains referred pain via shared second-order (or higher) neurons receiving input from both the affected viscera/deep structure AND a separate somatic dermatome — since the brain can't distinguish which source triggered the shared neuron, pain is perceived as coming from the somatic region instead.

Muscle Spindle & Golgi Tendon Organ — Fiber Types (6 questions)

Q81. Which afferent fiber type detects stretch of the muscle SPINDLE (intrafusal fibers)?

✓ **Correct: A. Ia afferent fiber**

The Ia afferent fiber detects stretch of the muscle spindle (intrafusal muscle fibers) — the sensory limb of the stretch (myotatic) reflex.

✗ **B is wrong:** Ib afferents detect Golgi tendon organ stretch/tension, not muscle spindle stretch — the paired, easily-confused fiber type.

PROF'S EMPHASIS: Ia = muscle spindle (stretch). Ib = Golgi tendon organ (tension). Same Roman numeral family, different letter, completely different receptor and reflex role — a favorite one-letter-swap trap.

Q82. Which afferent fiber type detects stretch/tension of the GOLGI TENDON ORGAN?

✓ **Correct: A. Ib afferent fiber**

The Ib afferent fiber detects tension sensed by the Golgi tendon organ, located at the muscle-tendon junction — providing the sensory signal for the inverse myotatic (autogenic inhibition) reflex.

Q83. Regarding the Achilles tendon reflex, which statement is correct?

✓ **Correct: A. The receptor is the muscle spindle (not the Golgi tendon organ, despite the tendon in the name)**

Despite the name, tendon-tap reflexes (like the Achilles reflex) are stretch reflexes — the tap stretches the muscle (via the tendon), and the muscle SPINDLE is the actual receptor detecting that stretch, triggering the monosynaptic reflex arc.

PROF'S EMPHASIS: Don't be fooled by the name 'Achilles TENDON reflex' — like all deep tendon/stretch reflexes, the actual RECEPTOR is the muscle spindle (detecting the tap-induced stretch), not the Golgi tendon organ. The 'tendon' in the name refers to where you TAP, not what senses the stretch.

Q84. Regarding nuclear bag and nuclear chain intrafusal fibers within the muscle spindle, which correctly describes their afferent innervation?

✓ **Correct: A. Nuclear bag fibers carry both static and dynamic responses (Ia primary afferent, with dynamic bag fibers giving the dynamic component); nuclear chain fibers carry mainly static response (Ia and II afferents)**

Nuclear bag fibers (particularly the dynamic bag1 subtype) contribute the dynamic (velocity-sensitive) component of the Ia response; nuclear chain fibers contribute more to the static (steady-length) response and receive both Ia and group II afferents — a commonly tested nuance of muscle spindle physiology.

PROF'S EMPHASIS: This is a genuinely fiddly, high-yield distinction: dynamic (bag1) fibers respond to the RATE of stretch (dynamic sensitivity), while static (bag2 and chain) fibers respond to the steady-state LENGTH (static sensitivity) — both bag and chain fibers are innervated by Ia afferents, but only chain fibers (plus static bag) additionally get group II afferents for sustained, static length information.

Q85. What is the function of GAMMA motor neurons in relation to the muscle spindle?

✓ **Correct: A. They maintain the sensitivity of the muscle spindle (by adjusting intrafusal fiber tension) during muscle shortening, so the spindle keeps detecting stretch even as the whole muscle contracts**

Gamma motor neurons innervate the intrafusal (spindle) muscle fibers, adjusting their tension so the spindle remains sensitive to stretch throughout the full range of muscle length — including during active muscle shortening, when the spindle would otherwise go slack and stop signaling.

✗ **D is wrong:** Preventing excessive force production is the GOLGI TENDON ORGAN's role (via the inverse myotatic reflex, Ib afferents), not the gamma motor neuron's role.

PROF'S EMPHASIS: Alpha motor neurons contract the WORKING (extrafusal) muscle. Gamma motor neurons contract the intrafusal fibers WITHIN the spindle, keeping the spindle taut/sensitive even as the whole muscle shortens — without gamma co-activation, the spindle would go slack and 'blind' during active contraction. This alpha-gamma coactivation concept is a favorite deeper-understanding question.

Q86. When dynamic gamma motor neurons are co-activated together with alpha motor neurons during a voluntary contraction, what is the effect...

✓ **Correct: A. MORE Ia impulses are generated (gamma coactivation keeps the spindle taut and responsive, preventing it from falling silent as the muscle shortens)**

Alpha-gamma coactivation means that as alpha motor neurons drive extrafusal contraction (shortening the whole muscle), gamma motor neurons simultaneously contract the intrafusal fibers, keeping spindle tension/sensitivity high — resulting in MORE, not fewer, Ia impulses than alpha activation alone would produce.

Free Nerve Ending + A-delta / C Fibers (4 questions)

Q87. Free nerve endings are found in the clefts between epidermal cells and convey which type of information?

✓ **Correct: A. Nociceptive (pain) and thermal information related to cellular injury**

Free nerve endings are the receptor structure for nociceptors, found at the skin and throughout body organs, conveying pain and thermal (temperature-related) information tied to tissue/cellular injury.

Q88. A pinprick to the skin activates which fiber type FIRST, producing fast/first pain?

✓ **Correct: A. A-delta fibers**

A pinprick classically activates A-delta fibers, producing fast/first pain — sharp, well-localized, immediate.

PROF'S EMPHASIS: Pinprick = fast/sharp/first pain = A-delta. Dull ache that follows (e.g., after a muscle pull) = slow/second pain = C fiber. Your professor's slide uses EXACTLY these two example scenarios (pinprick vs. muscle-pull ache) — recognize them instantly.

Q89. A dull, persistent ache that follows a muscle pull results from activation of which fiber type?

✓ **Correct: A. C fibers**

Dull, persistent, poorly-localized ache (like that following a muscle strain) is the signature of C-fiber-mediated slow/second pain.

Q90. Which chemical mediators are associated with activating/sensitizing free nerve ending nociceptors, per your lecture?

✓ **Correct: A. Histamine, substance P, prostaglandin, and K+**

Your lecture specifically diagrams histamine, substance P, prostaglandin, and K⁺ as chemical mediators acting on free nerve ending nociceptors, generating a receptor potential that (if threshold is reached) becomes an action potential perceived as pain.

PROF'S EMPHASIS: This is the same 'inflammatory soup' concept tested elsewhere: histamine, substance P, prostaglandin, K⁺ (plus bradykinin/H⁺ in fuller lists) sensitize/activate nociceptors — memorize this as a set since 'which of these does NOT activate nociceptors' is a common negatively-phrased question format.

Somatosensory Tracts (7 questions)

Q91. Which THREE major ascending sensory pathways carry general somatic afferent (GSA) information (touch, pressure, vibration, pain,...)

✓ **Correct: A. (1) Anterolateral system (anterior + lateral spinothalamic tracts), (2) Dorsal column-medial lemniscus system (fasciculus gracilis + cuneatus), (3) Somatosensory-to-cerebellum pathways (ventral/dorsal spinocerebellar, cuneocerebellar)**

Your lecture organizes GSA ascending pathways into three systems: the anterolateral system (anterior + lateral spinothalamic tracts, carrying pain/temperature/crude touch), the dorsal column-medial lemniscus system (fasciculus gracilis + cuneatus, carrying fine touch/vibration/conscious proprioception), and somatosensory-to-cerebellum pathways (ventral/dorsal spinocerebellar and cuneocerebellar tracts, carrying unconscious proprioception).

PROF'S EMPHASIS: Three systems, three jobs: Anterolateral (pain/temp/crude touch, **CROSSED**). DCML (fine touch/proprioception/vibration, crosses **LATE** in the medulla). Spinocerebellar (unconscious proprioception **TO** the cerebellum, mostly uncrossed). Sort every somatosensory tract question into one of these three buckets first.

Q92. Which tract carries CONSCIOUS proprioception, fine/discriminative touch, and vibration sense?

✓ **Correct: A. Dorsal column-medial lemniscus pathway (fasciculus gracilis + cuneatus)**

The DCML pathway is the one that reaches conscious awareness (via VPL thalamus and primary somatosensory cortex) — the spinocerebellar tracts, by contrast, carry proprioceptive information that never reaches consciousness.

Q93. Which tract carries pain and temperature sensation, and where does it cross the midline?

✓ **Correct: A. Lateral spinothalamic tract, crossing within 1-2 segments of entering the cord (via the anterior white commissure)**

The lateral spinothalamic tract carries pain and temperature; its second-order fibers cross the midline early, within about 1-2 spinal segments of entering the cord, via the anterior white commissure — much earlier than the dorsal column pathway crosses (which happens up in the medulla).

Q94. Which somatosensory tract synapses at Clarke's nucleus, carrying UNCONSCIOUS proprioception from the lower limb to the cerebellum?

✓ **Correct: A. Dorsal spinocerebellar tract**

The dorsal spinocerebellar tract's second-order neurons ARE the neurons of Clarke's nucleus (present roughly C8-L2/3), carrying unconscious proprioceptive information from the lower limb to the ipsilateral cerebellum.

PROF'S EMPHASIS: Clarke's nucleus exists only roughly C8-L2/3 — a level-restricted structure, which is exactly the kind of 'where does this apply' nuance that turns a simple fact into a harder integrative question.

Q95. Regarding touch sensation specifically, which statement correctly identifies the SECOND-order neuron's synapse location?

✓ **Correct: A. The second-order neuron for touch (via the dorsal column pathway) synapses in the cuneate (or gracile) nucleus of the brainstem/medulla**

For touch traveling via the DCML pathway, the first-order neuron (DRG) ascends ipsilaterally in the dorsal column and synapses on the second-order neuron in the cuneate nucleus (upper body) or gracile nucleus (lower body) of the medulla — only THEN does the pathway cross (as internal arcuate fibers) before reaching the thalamus (3rd-order neuron) and finally the cortex (4th-order neuron).

✗ **B is wrong:** The THIRD-order neuron synapses in the thalamus, not the second — an easy off-by-one error in the neuron-order chain.

PROF'S EMPHASIS: DCML neuron order: 1st-order (DRG, ascends ipsilaterally in the dorsal column) → 2nd-order (synapses in gracile/cuneate nucleus, THEN crosses as internal arcuate fibers) → 3rd-order (VPL thalamus) → 4th-order (primary somatosensory cortex). Touch's 2nd-order synapse in the cuneate/gracile nucleus, not the thalamus, is a commonly tested specific fact.

Q96. For touch sensation, in which structure does the FOURTH-order neuron reside?

✓ **Correct: A. Primary somatosensory (somesthetic) area of the cerebral cortex**

The full touch pathway chain: 1st-order (DRG) → 2nd-order (cuneate/gracile nucleus) → 3rd-order (VPL thalamus) → 4th-order neuron, located in the primary somatosensory cortex itself.

Q97. Which tract carries pain and temperature information from the FACE (via the spinal trigeminal nucleus), functionally analogous to the...

✓ **Correct: A. Ventral trigeminothalamic tract**

The spinal trigeminal nucleus (receiving pain/temperature/crude touch input from CN V, and smaller GSA contributions from CN VII, IX, X) relays via the ventral trigeminothalamic tract to the thalamus — the facial/head equivalent of the body's lateral spinothalamic tract.

PROF'S EMPHASIS: Body pain/temp = (lateral) spinothalamic tract. Face pain/temp = (ventral) trigeminothalamic tract, via the spinal trigeminal nucleus. Structurally and functionally parallel systems for two different body regions — a favorite 'body vs. face equivalent' pairing question.

Taste (Gustatory) Pathway (5 questions)

Q98. What is the correct taste (gustatory) pathway from receptor to cortex?

✓ **Correct: A. Taste receptor cells (CN VII, IX, X) → Geniculate/Petrosal/Inferior(nodose) ganglia → Solitary nucleus (rostral/upper part) → Solitary tract → VPM of thalamus → Gustatory area of cerebral cortex (area 43, near the midline)**

Taste afferents from three different cranial nerves (VII, IX, X) converge on the rostral (gustatory) part of the solitary nucleus after passing through their respective ganglia, then ascend via the solitary tract to the VPM (not VPL) nucleus of the thalamus, finally reaching the gustatory cortex (area 43, near the midline/insula region).

✗ **C is wrong:** This describes the **FACIAL PAIN/TEMPERATURE** pathway (trigeminal, spinal trigeminal nucleus, VPM, somatosensory cortex) — a structurally similar-sounding but functionally distinct pathway from taste, easy to confuse since both use VPM.

PROF'S EMPHASIS: Full chain, exactly as your professor draws it: Taste bud → CN VII/IX/X → Geniculate/Petrosal/Inferior ganglion → Solitary nucleus (rostral part specifically) → Solitary tract → VPM thalamus (NOT VPL — that's for body touch) → Gustatory cortex, area 43. Every single link here has shown up as its own separate exam question before.

Q99. Which ganglia house the first-order taste neurons for CN VII, IX, and X respectively?

✓ **Correct: A. CN VII = Geniculate ganglion; CN IX = Petrosal ganglion; CN X = Inferior (nodose) ganglion**

Each of the three taste-carrying cranial nerves has its own dedicated sensory ganglion: CN VII uses the geniculate ganglion, CN IX uses the petrosal ganglion, and CN X uses the inferior (nodose) ganglion.

PROF'S EMPHASIS: Geniculate=VII, Petrosal=IX, Nodose(inferior)=X — three nerves, three ganglia, three tongue/pharynx regions (anterior 2/3, posterior 1/3, epiglottis) as a matched trio. Any of these three pairings could be tested in isolation.

Q100. Which part of the solitary nucleus specifically handles TASTE information, as opposed to general visceral/cardiopulmonary information?

✓ **Correct: A. The rostral (upper) part of the solitary nucleus**

The solitary nucleus is functionally subdivided: its **ROSTRAL** part processes taste (gustatory) information specifically, while its more caudal portions handle general visceral afferent (cardiopulmonary, GI) information.

Q101. What is the medical term for an unpleasant or distorted taste sensation?

✓ **Correct: A. Dysgeusia**

Dysgeusia specifically means distorted or unpleasant taste — distinct from the other similar-sounding 'dys-' terms for swallowing, voice, articulation, or coordination.

PROF'S EMPHASIS: Word bank: -phagia=swallowing, -phonia=voice, -geusia=TASTE, -arthria=speech articulation, -metria=cerebellar movement accuracy, -osmia=smell. Five to six similar 'dys-' terms are a classic examiner trap set — know the full glossary, not just one term in isolation.

Q102. Which stem cell type regenerates sensory receptor cells in BOTH the tongue (taste buds) and the olfactory epithelium?

✓ **Correct: A. Basal cells**

Basal cells serve as the shared regenerative stem cell population for both taste bud cells and olfactory receptor neurons — a unifying fact across two different special senses, tested specifically in your lecture.